

The WORKHOUSE Program

Exact $SU(N)$ Flux-Band Spectra, the Fixed-Spacing Wilson Construction,
and a Verified Register of What the Continuum Limit Still Requires

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Abstract

This is the master reference edition of a four-year program on the strong-coupling spectral structure of Hamiltonian $SU(N)$ lattice gauge theory and on the obligations that separate that structure from a continuum Yang–Mills mass gap. It is organized so that its two halves can be read, and used, independently.

Part I is a closed body of exact mathematics. A rank-one cellular subspace — the kernel of the plaquette boundary map — carries the strong-coupling band, and its hopping amplitude is the rational function

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)},$$

positive for $N \geq 3$, vanishing identically at $N = 2$. The four shared-link channel weights that produce it are order-two Weingarten values rather than an isotropy assumption, so the second-order theory rests on representation theory alone. At fourth order the effective operator splits into Hodge terms and one geometric term coupling the carrier to its complement, with the off-axis coefficient decided by derivation across three independent implementations. The rank grading is exact: every even order of the one-plaquette splitting carries the same single power of N , with coefficients $c_2 = -4$, $c_4 = -32$, $c_6 = -1748/3$, $c_8 = -123332/9$, $c_{10} = -49593808/135$, and a four-channel cancellation that sums to zero at every even order from four to ten; while the odd orders are determinant families — for even N every odd order vanishes identically, and for odd N the first surviving odd order is $N - 2$, with a closed-form single vertex. At sixth order, a rational band shape is proved to survive every local direct term. Of the statements in Part I, none is conditional on an open estimate.

Part II states the construction at fixed lattice spacing: an actual infinite-volume Wilson transfer limit, a complete physical odd band isomorphic to $\ell^2(\mathbb{Z}^3) \otimes \mathbb{C}^3$, and a literal plaquette-source frame that is onto that entire band, on an explicit interval of the coupling.

Part III is the part written for whoever works on this next. It states six obstructions between the fixed-spacing construction and the continuum limit — five of them with explicit finite counterexamples, four of them having killed arguments this program had written down as complete — of which the decisive one is that the interval where the construction is proved and the trajectory along which the continuum is taken are disjoint sets of couplings. It then gives the refutation register: 4 falsified claims with what falsified each, and the mechanisms this program advanced and withdrew. It closes with an obligation register of eight numbered problems and the repository’s full route record — 111 routes, of which 64 are complete, 20 live, 12 untried and 15 dead, each dead one carrying the reason it died.

No claim is made here about the existence of continuum Yang–Mills theory or about a continuum mass gap. §14 is the argument that no such claim currently follows, and §16 is the list of what would have to change.

Every count quoted above is generated from the repository’s ledgers rather than typed, and every statement carries its machine-verification tier and the command that re-establishes it. Currently: 439 Lean 4 theorems with 0 occurrences of `sorry`, 143 registered analytic results, and 54 verified evidence edges to published work across 34 obtained papers.

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1 Introduction

1.1 What is actually here

Strong-coupling expansions of Hamiltonian lattice gauge theory are old. What is not old is the object this program organizes them around.

Write the cellular chain complex of the cubic lattice as $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$, with C_2 spanned by plaquettes. The subspace

$$Z_2 = \ker \partial_2 \subset C_2 \tag{1.1}$$

— the closed plaquette chains — is the carrier of the strong-coupling band. It is a homological object, fixed before any perturbation theory is done, and the entire content of Parts **I** and **II** is a description of what the Kogut–Susskind Hamiltonian does to it, order by order and rank by rank.

Three things follow from taking that object seriously, and they are the three claims of this document.

First: the coefficients are exact rational functions of the rank. Not asymptotic in N , not fitted, not computed rank by rank and extrapolated. The second-order hopping amplitude is

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}, \quad t_2 = 0, \quad t_3 = \frac{5}{612}, \tag{1.2}$$

and the four channel weights behind it are order-two Weingarten values, so (1.2) imports nothing from any expansion. The factor $N^2 - 4$ is not a coincidence of small rank: it is why $SU(2)$ has no band at this order at all. The same discipline runs through fourth order, through the rank grading

of §7, and into a symbolic engine that works over $\mathbb{Q}(N)$ with N an indeterminate rather than a number.

Second: the rank grading has exact structure, and it comes from cancellation. For the C -parity splitting of the one-plaquette diagonal, every even order carries the same single power of N once the coupling is written in the planar variable, and the leading coefficients of the four contributing intermediate channels are in the ratio $(-4, +2, +1, +1)$ and *sum to zero*. The grading is a cancellation theorem, not a power count. Complementarily, the odd orders vanish identically for even N , and for odd N the first surviving one is order $N - 2$, with a closed-form vertex. Neither statement is visible from any single rank.

Third: the construction at fixed lattice spacing is complete, and it does not reach the continuum. §12 states an actual infinite-volume Wilson transfer limit with a complete physical odd band and a literal plaquette-source frame that is onto it. §14 states why that is not a continuum mass gap, in six independent ways, five of them with finite counterexamples. The most important is structural rather than technical: the theorem holds on $|u| \leq u_*/(10\,022\,400\,000N)$ and the continuum trajectory runs to $u \rightarrow \infty$, so the two are disjoint and no sharpening of the constant changes that.

1.2 Why the document is shaped this way

Alex, the maintainer of this program, asked for a paper that would serve as the base for further work on the mass gap rather than as a report of work completed. That request has consequences for the form.

It means that failure is content. §14 and §15 exist because a route this program has already closed is more valuable to the next attempt than a route it has already opened, and because at least four arguments were written down here as complete and then found not to be. It means the route register in §16 reproduces 15 dead routes at length, including their measured costs, rather than listing only the 20 live ones. And it means every count in the prose is a macro generated from the repository's ledgers, so that a stale number is a build failure rather than a quiet misstatement.

It also means the document distinguishes, everywhere and visibly, three things that are routinely conflated: what a statement mathematically establishes, what artefact backs it, and what a machine checks. A result can be fully proved analytically and carry the weakest machine badge. A Lean theorem can be checked by the kernel and prove something trivial. This program has made both errors and the notation of §2 is the correction.

1.3 Relation to the literature

The external comparisons are recorded as an evidence map rather than a citation list: 108 indexed papers, 152 recorded relations, of which 54 are verified against a named claim here. A published paper is not authority in this program — it is unchecked until something checks it, exactly like an internal document. What makes an external agreement evidence is that it was produced without knowledge of this program, and §11 reports the comparisons that hold and, separately, the two that contradict.

1.4 What is claimed, and what is not

Claimed: the contents of Part I, unconditionally, at the tiers recorded; and the contents of Part II on the stated interval and under the stated hypotheses.

Not claimed: the existence of continuum four-dimensional Yang–Mills theory; a continuum mass

gap; Poincaré covariance; restoration of rotational invariance from the hypercubic lattice; or the construction of a continuum measure. None of these has an argument in this document, and §14 explains for each of the six obstructions why the present machinery does not supply one.

2 How to use this document

This document has two audiences and it is worth saying which paragraph is for whom.

A reader who wants the mathematics should read Part I, which is self-contained and compiles on its own as a separate paper (`paper/master/workhouse_band_paper.tex`). Nothing in Part I depends on anything later, and nothing in it is conditional on an open estimate.

A reader who intends to *work* on the continuum problem should read §14, §15 and §16 first, in that order, and only then come back for the mathematics. The reason is economic. This program has spent four years discovering that most of the obvious routes to the continuum limit are closed, and the record of *which* routes are closed and *why* is worth more than the record of what is open. Four sessions of one earlier effort were spent on a single route that a static argument had already shown could not decide the question it was aimed at; that experience is why the route register in §16 exists and why dead routes are reproduced in it at full length rather than deleted.

2.1 What the badges mean

Every statement in this document carries a machine-verification badge.

- [T0 Lean] — Lean 4 compiles the statement from explicit definitions with no `sorry`, using only `propext`, `Classical.choice` and `Quot.sound`. Nothing written in prose can weaken a T0 badge.
- [T1 exact] — the statement is re-derived symbolically, in exact arithmetic, from stated definitions.
- [T2 numeric] — floating-point agreement within a tolerance that the check itself prints.
- [T3 analytic] — no single repository check covers the whole statement.

The badge is *independent* of mathematical status. A theorem with a complete analytic proof and no formalization carries [T3 analytic]: the badge says what the machine checks, not what the mathematics establishes. This is the one convention a reader must internalize to read the document correctly, because the opposite reading — T3 as a synonym for “unproved” — makes most of Part III look weaker than it is, and the reading of T0 as a synonym for “important” makes it look stronger. Both errors have been made in this program’s own internal documents and corrected.

Where a statement’s scope is narrower than its headline suggests, a **Does not assert** paragraph follows it. Those paragraphs are not hedging. They are the part of each result that took longest to get right, and several of them exist because an earlier version of the claim was broader and was refuted.

2.2 If you are the next agent on this program

The following is an operating procedure, not advice. It is written for the case — now the normal case — in which the next piece of work on this program is done by an automated agent with a fresh context window.

1. **Do not trust this document over the ledgers.** Every number quoted in the prose here is generated from `ledger/*.yaml` by `paper/master/generate_apparatus.py`. If the ledger and this document disagree, the ledger is right and this document is stale; regenerate it. The counts in §16, §11 and Appendix B are macros, not typed literals, precisely so that this failure mode is visible rather than silent.
2. **Establish the state before choosing a target.** Run `workhouse brief --startup`, then `workhouse brief <gap> --json` for the gap you mean to work on, and retain the snapshot. A saved graph in a fresh clone reports freshness `unknown` until local generation provenance exists; saved briefing mode never silently rebuilds it.
3. **Read the dead routes for your gap before starting.** `workhouse why <gap>` prints them. §16 reproduces them here. Of 111 recorded routes across all gaps, 15 are dead, and several died only after substantial expenditure with a measured cost recorded in the ledger. A route marked `cannot_decide` is not merely unpromising: it is structurally incapable of resolving the claim it was aimed at, and re-running it will produce output that answers a different question.
4. **Do not aim at the continuum limit directly.** §14 states six independent obstructions, five of them with explicit finite counterexamples. Each is a constraint on what a correct argument can look like, and an argument that does not explicitly clear all six is not a candidate. In particular, Break 5 (§14.5) shows that the regime in which this program’s strongest construction holds and the regime in which the continuum limit lives are disjoint sets of couplings. That is a statement about the shape of any possible proof, not a gap in the present one.
5. **Pick a numbered Problem from §16**, not a topic. Each Problem states its available hypotheses, what it unblocks downstream, and the specific next computation. Problems are ordered by the ledger’s research judgement, which is not a claim that earlier ones are prerequisites for later ones.
6. **Record the attempt, including its failure.** A route that dies is a result. Set its state in `ledger/gaps.yaml`, name what closed it in `closed_by`, and if the instrument could not have reached the claim, say so in `cannot_decide`. The working agreement in `CLAUDE.md` is the binding version of this instruction; this paragraph only explains why it is there.

2.3 What would falsify the contents of this document

Three things, in decreasing order of how much they would cost.

First, an error in the Haar or Weingarten input to §5 would propagate to every order and to both the fourth-order band and its rank grading. That input is now representation theory rather than assumption (§5), and it is the load-bearing algebraic fact of Part I.

Second, a demonstration that the plaquette source family of §12 is not total in the physical complement would empty the fixed-spacing band theorem of physical content without making it false — this is exactly the failure mode that Break 4 (§14.4) exhibits in a three-dimensional model, and totality is therefore asserted in that theorem as a proved conclusion rather than assumed.

Third, and least costly, a published result reproducing any of the exact coefficients of §7 or §6 would remove this program’s claim to have derived them first without touching their correctness. §11 records the searches already performed against 108 indexed papers.

Part I

Exact spectral structure at strong coupling

3 Setting and conventions

3.1 The Hamiltonian

Work on the cubic lattice $\mathbb{Z}^3$ with gauge group $G = \mathrm{SU}(N)$, one group element U_ℓ per oriented link. The Kogut–Susskind Hamiltonian is

$$H = H_0 + V, \quad H_0 = \frac{1}{2} \sum_{\ell} E_{\ell}^2, \quad V = -u W, \quad (3.1)$$

where E_{ℓ}^2 is the quadratic Casimir of the left-invariant electric field on link ℓ , and W is the sum of plaquette characters in both orientations,

$$W = \sum_p (\chi_p + \bar{\chi}_p), \quad \chi_p = \mathrm{Tr} \prod_{\ell \in \partial p} U_{\ell}. \quad (3.2)$$

Strong-coupling perturbation theory is the expansion in u around the electric vacuum, organized as degenerate perturbation theory — a des Cloiseaux effective operator — on the sector of interest.

3.2 The coupling, and one erratum that matters

The canonical coupling variable throughout is

$$u = \frac{\beta_N}{2N}, \quad (3.3)$$

whose $\mathrm{SU}(3)$ specialization is $u = \beta_3/6$.

A warning, because it has corrupted transcriptions of this material before. An archived line reading $Y = 2\beta/3 = 4u$ is a definition-label erratum, not a change of variable. The coefficients printed alongside it were *already* in the canonical coordinate (3.3). Applying a 4^r rescaling at order r to “correct” them corrupts every order, and the repository carries an explicit check that the printed towers reproduce their canonical values verbatim.

For rank-uniform statements the natural variable is the planar one,

$$\tau = \frac{2u}{N^2} = \frac{\beta_N}{N^3}, \quad (3.4)$$

in which the grading theorems of §7 take their simplest form: every even order of the one-plaquette splitting then carries the same single power of N .

3.3 Two projections that must not be conflated

Two distinct projections appear throughout, and several contradictions in this program’s own register were naming collisions between them.

The *band* projection is onto the degenerate strong-coupling subspace carrying the excitation of interest — for the flux band, the cellular subspace of §4. Coefficients anchored to it are band-kernel quantities.

The *physical* projection is onto the gauge-invariant, vacuum-subtracted spectrum at a definite momentum. Coefficients anchored to it are physical quantities.

These are related by translation-local scalar shifts, and such a shift changes neither the centered operator, nor its eigenvectors, nor the bandwidth. It does change the scalar. Consequently the band-kernel anchor and the physical Γ -point coefficient are two differently anchored coordinates, not two competing estimates of one quantity, and they are written $q_{\text{band}}^{(4)}$ and $m_{\Gamma}^{(4)}$ respectively. Writing either as “ m_4 ” regenerates a contradiction that does not exist, and the repository’s search tooling refuses the name for that reason.

3.4 Exactness

Every coefficient displayed in Part I is an exact rational, or an exact rational function of N . Where a quantity is known here only numerically it carries an explicit numerical marker in the source registry and is never displayed as though exact. The single most dangerous defect available in this material is a float that reads as exact, and the boundary between the two is machine-guarded rather than maintained by care.

4 The homological carrier

4.1 Where the flat band comes from

Project the Kogut–Susskind Hamiltonian (3.1) onto the fundamental, charge-odd, one-plaquette manifold. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link. The projected problem is therefore a three-orbital Bloch Hamiltonian, and it looks at first like a tight-binding model.

It is not an engineered one. The signs of the matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and non-abelian group integration generates, exactly, the oriented edge–face boundary Gram operator

$$B(k)B(k)^\dagger, \quad B = \partial_2^\dagger, \quad (4.1)$$

with the coefficient of §5 in front of it. The structure of the result is a clean separation:

$$\boxed{\text{colour dynamics} \longrightarrow t_N, \quad \text{cellular geometry} \longrightarrow BB^\dagger.} \quad (4.2)$$

Representation theory fixes the amplitude; the cubical chain complex fixes the kernel and the spectrum. Neither borrows from the other.

4.2 The projected spectrum

Theorem 4.1 (Second-order projected spectrum). *The projected second-order spectrum consists of one momentum-independent level together with a doubly degenerate branch shifted by $t_N u^2 q(k)$, where*

$$q(k) = 4 \sum_{j=1}^3 \sin^2(k_j/2). \quad (4.3)$$

The flat level is exact and it is exact for a reason: the flat carrier is $\ker \partial_2$.

Theorem 4.2 (The carrier and its multiplicity). *Elementary cube boundaries are compact localized states of the carrier; three wrapping sheets complete it on the torus. For $L \geq 3$ the exact multiplicity is*

$$(L^3 - 1) + 3 = L^3 + 2. \quad (4.4)$$

Graph id: RESULT:UNIVERSAL_CELLULAR_HODGE_SPECTRUM

The two contributions are structurally different and the distinction matters later. The $L^3 - 1$ cube boundaries are local and finitely supported. The three sheets are not: they wrap the torus, and they are the entire zero-momentum content of the carrier. §12 returns to the consequence — that no finitely supported exactly-flat local operator can see the Γ point at all.

4.3 Third order, and the coefficients that survive

For $SU(3)$, exact third-order perturbation theory stays inside the same boundary-factorized algebra. It does not generate a new operator structure; it renormalizes the coefficient in front of the same $BB^\dagger$:

$$H_{\text{eff},-} = E_{\text{flat}}(u) I + \left(\frac{5}{612} u^2 + \frac{1975}{124848} u^3 \right) BB^\dagger + O(u^4), \quad (4.5)$$

with

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306} u^2 - \frac{109151}{249696} u^3. \quad (4.6)$$

That (4.5) remains boundary-factorized through third order is the non-trivial part. It is what makes the fourth order the first place where the carrier can couple to its complement, and §6 is about exactly that coupling.

4.4 What is and is not novel here

This program does not claim that incidence flatness, cellular compact localized states, or homological counting of zero modes are new. All three have a substantial prior literature — line-graph constructions, compact localized states, the necessity of non-contractible states on periodic systems, rectangular factorizations and index arguments, and closely related closed-surface structures in two-form spin liquids — and a recent independent construction treats face-graph and higher-cell flat bands directly.

What is claimed is that the flatness here is *dynamical*: it is produced by exact strong-coupling matrix elements of non-abelian $SU(N)$ Hamiltonian lattice gauge theory, with the amplitude fixed by group representation theory and the kernel fixed by the cubical chain complex, as in (4.2). The flat band is not put in; it comes out.

Scope 4.3. Theorem 4.1 and (4.5) are finite-volume, finite-order statements about a projected flux sector. They are not a continuum glueball theorem and not a mass-gap theorem, and the source manuscript says so in its own abstract.

5 Second order at every rank

This section is the algebraic base of the whole program. Everything in it is proof-checked, and its conclusion — that the band exists for $N \geq 3$ and not for $N = 2$, with an exact amplitude — follows from representation theory with no input from any expansion.

5.1 The shared-link weights are Weingarten values

Two plaquettes contributing at second order either share a link or do not. The amplitude for the shared-link case was for a long time taken in this program with an isotropy premise: that the four contributing representation channels carry weights in a fixed ratio. That premise is not needed. It is a consequence of the order-two Weingarten values.

Proposition 5.1 (The shared-link weights, from representation theory alone). *Let two plaquettes share exactly one link. Then:*

1. *The six non-shared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar. So the two-plaquette amplitude reduces to $\text{Tr}(AU) \text{Tr}(BU^{\pm 1})$ with A and B independent Haar, and integrating them out leaves a pure degree-(2,2) moment of the shared link U .*
2. *Two such moments settle both families. With $M_{\text{direct}} = N^2$ and $M_{\text{cross}} = N$, the like family splits as $(N+1)/(2N)$ and $(N-1)/(2N)$; and the singlet component of $U_{ij}\overline{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1, giving $1/N^2$.*

All four weights are exactly d_R/N^2 , where d_R is the dimension of the channel.

Graph id: theshared-linkweightsareWeingarten,notanisotropyassumption Reproduce: workhouse verify -only 'the shared-link weights are Weingarten, not an isotropy assumption'

[T1 exact]. The relevant Weingarten pair is the inverse of the S_2 Gram matrix and the index sums are explicit, so the check imports nothing from the corpus. Registered as the discharge of the gap that had carried this premise as the one unproved input of the second-order chain; it was opened and closed on the same day, 28 August 2026, once the question was asked in the right form.

5.2 The four channels in closed form

Write $C_F = (N^2 - 1)/(2N)$ for the fundamental Casimir. Each channel R contributes a resolvent weight $-(d_R/N^2)/(C_F + \frac{1}{2}\Delta C_2(R))$, and all four evaluate in closed form.

Theorem 5.2 (Per-channel resolvent weights).

$$w_{\mathbf{1}} = -\frac{1/N^2}{C_F} = -\frac{2}{N(N^2 - 1)}, \quad (5.1)$$

$$w_{\text{Adj}} = -\frac{(N^2 - 1)/N^2}{C_F + N/2} = -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, \quad (5.2)$$

$$w_{\text{Sym}} = -\frac{N(N+1)/2N^2}{C_F + \frac{(N-1)(N+2)}{2N}} = -\frac{N+1}{(N-1)(2N+3)}, \quad (5.3)$$

$$w_{\Lambda} = -\frac{N(N-1)/2N^2}{C_F + \frac{(N+1)(N-2)}{2N}} = -\frac{N-1}{(N+1)(2N-3)}. \quad (5.4)$$

Graph id: LEAN:resolventWeight_singlet,LEAN:resolventWeight_adjoint,LEAN:resolventWeight_sym,LEAN:resolventWeight_antisym

[T0 Lean] — all four, at lean/Workhouse/Basic.lean:72,80,89,104. Reproduce with make lean.

5.3 The two family sums

The channels group into a *mixed* family $\{\mathbf{1}, \text{Adj}\}$ and a *like* family $\{\Lambda^2 F, \text{Sym}^2 F\}$, and these are the antiparallel and parallel channel sums.

Theorem 5.3 (Family sums).

$$A_N = w_{\mathbf{1}} + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad B_N = w_{\text{Sym}} + w_{\Lambda} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (5.5)$$

Graph id: LEAN:mixed_family_sum, LEAN:like_family_sum

[T0 Lean], at `lean/Workhouse/Basic.lean:118,129`. The mixed sum is obtained from the dimension and Casimir table rather than by transcription from any source document, which is the point: the family sums are re-derived, not quoted.

5.4 The hopping amplitude

Theorem 5.4 (The all-rank second-order hopping amplitude).

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}. \quad (5.6)$$

It is strictly positive for every $N \geq 3$, and

$$t_2 = 0, \quad t_3 = \frac{5}{612}. \quad (5.7)$$

Graph id: CONST:t_N Reproduce: workhouse why t_N

[T0 Lean] for the rank law and for both special values (`lean/Workhouse/Basic.lean:rank_law_numerator`, `hopping_two`, `hopping_three`). Clearing denominators, the channel difference has numerator exactly $2N(N^2 - 4)$.

Two consequences deserve to be stated separately, because both are structural rather than numerical.

Corollary 5.5 (SU(2) has no band at this order). *The factor $N^2 - 4$ in (5.6) vanishes at $N = 2$. The exclusion of SU(2) is therefore not a smallness but an identity: the parallel and antiparallel channel sums coincide exactly at rank two.*

Graph id: LEAN:hopping_two

Corollary 5.6 (The planar approach is from below). *With $D = (N^2 - 1)(2N^2 - 1)(4N^2 - 9)$,*

$$D - 4N^3 \cdot 2N(N^2 - 4) = 2N^4 + 31N^2 - 9 > 0 \quad (N \geq 1), \quad (5.8)$$

so $4N^3 t_N = 1 - (2N^4 + 31N^2 - 9)/D < 1$. Hence $t_N < 1/(4N^3)$ for every N , and $t_N \sim 1/(4N^3)$: the N^{-3} cancellation is exact and what it leaves is a strictly positive deficit.

Graph id: LEAN:hopping_deficit_numerator

[T0 Lean], at `lean/Workhouse/Basic.lean:48`. Corollary 5.6 is the first appearance of the rank grading that §7 develops: the leading power of N is fixed by a cancellation, and the surviving term is a polynomial one can write down.

5.5 Independent corroboration of the channel weights

The weights of Proposition 5.1 have an external provenance, from work written without knowledge of this program.

An efficient-truncation study of $SU(N_c)$ lattice gauge theory for quantum simulation [21] gives, in its Appendix D, the exact finite-rank plaquette matrix element

$$|M_\rho| = \sqrt{\frac{\dim \rho}{\dim A \dim R}}. \quad (5.9)$$

Taking both factors fundamental gives $|M_\rho|^2 = d_\rho/N^2$, which is *identically* the numerator of the weight formula used here. The four shared-link channel weights, and with them A_N , B_N and $t_N = B_N - A_N$, therefore rest on a matrix element an independent group wrote down for an unrelated purpose. That truncation is also the first published one retaining all four shared-link irreducible representations.

The agreement is evidence rather than bookkeeping precisely because neither side knew of the other. It does not make t_N published: the rank law (5.6), the vanishing at $N = 2$, and the deficit of Corollary 5.6 are assembled here.

A caution attaches in the other direction. The leading-large- N_c qutrit plaquette Hamiltonian [20] explicitly decouples the charge-odd combination, so the sector in which t_N lives is inert in that model — effectively $t_3 = 0$ at leading order in large N_c , against the finite- N channel-complete value 5/612. That paper is recorded in this program’s index as not yet obtained, and the statement here is taken from imported bridge documents rather than checked against the paper itself.

5.6 What this section does not establish

Does not assert. Convergence of the strong-coupling series; anything about the spectrum at finite u beyond second order; and nothing about $SU(2)$ at higher orders, where Corollary 5.5 does not apply. Theorem 5.4 is a statement about one coefficient of a formal expansion, established exactly.

6 Fourth order, and how a disputed coefficient was decided

Fourth order is where the carrier first couples to its complement, and it is where this program spent the largest single block of its effort. The mathematics is one theorem and one number. The number was disputed for months between two irreconcilable values, and the way it was settled is, in this author’s view, the more instructive half of the story.

6.1 The structure

The fourth-order effective operator splits into Hodge terms and one geometric term coupling the carrier to its complement. Restricted to range-one operator structures, the shape content collapses: each such structure contributes a fixed shape coefficient, and every one of them gives $B = D = 0$ identically. The tier collapse is simply what range-one looks like.

Theorem 6.1 (Fourth-order support, at $SU(3)$). *The recorded fourth-order kernel has support $\{I, U, S, S^2, R\}$, and the shape data closes in that span. Solved over the whole Brillouin zone in exact rationals rather than fitted, the shape table is determined by three signed amplitudes:*

$$C_{\text{shp}} = -\frac{5}{96} - u - \frac{1}{2}(\rho + \pi), \quad u_2 = 2u, \quad \nu = -\left(\frac{5}{48} + 4u\right), \quad (6.1)$$

where u is the two-hop chain cumulant.

Graph id: RESULT:TIER_COLLAPSE_ACTUAL_H4_SUPPORT, RESULT:TIER_COLLAPSE_IS_R_DEGREE

Scope 6.2. “The recorded kernel” means the 189-record historical kernel, which is $N = 3$. The operator splitting is an $SU(3)$ statement; it is not established at general rank, and framing it as an $SU(N)$ identity would be an overreach. The rank-uniform statements of this program are the coefficient laws of §5 and §7, not this splitting.

Clearing the $1/q$ makes the shape ansatz a linear system over Laurent polynomials in e^{ik_j} , which is why the whole zone can be solved exactly instead of sampled.

There is a second, equivalent presentation. In the Hodge form the two-hop weight is absorbed, $\tilde{\pi} = \pi + 2u$, and

$$C_{\text{shp}} = -\frac{5}{96} - \frac{1}{2}(\rho + \tilde{\pi}), \quad (6.2)$$

so that u drops out of C entirely and exactly two shared-link amplitudes remain. Equations (6.1) and (6.2) are the same statement. They must not be combined — writing u and $\tilde{\pi}$ in the same expression double-counts the two-hop weight, and that error has been made.

6.2 The dispute

Two independently recorded kernels — a historical pipeline from June 2026 and a later cold dump — gave different values for the off-axis shape coefficient C_{shp} . Both were exact rationals. Neither could be promoted, because the repository’s governing rule is that a dispute closes by derivation or not at all: code may not pick a side, average them, or prefer whichever rational looks more authoritative.

Comparing the two kernels record by record localized the disagreement rather than resolving it. Both have the same six weight orbits and the same normalized shape table; both satisfy the same two relations above. What differs is the skeleton scale and the *signs* of two amplitudes. One scale factor explains 144 of the 189 records to 2×10^{-12} : the rotation bulk is structurally shared, and the dispute is three amplitudes, not two kernels.

6.3 How it was settled

By building a third implementation that shares no primitive with either, and by recovering the historical pipeline’s own intermediate ledger.

The third implementation is a Wilson-loop calculus written from scratch: the free Hamiltonian by the Fierz identity as a rewiring of index ports — a like pair swaps wires, an unlike pair is cut and crossed, which is unitarity; Haar integrals by the $U(N)$ Weingarten function as the Gram pseudoinverse on S_n , with $\det U = 1$ inserted on virtual slots for the determinant families; and the resolvent by per-link irreducible projectors, so that every component is an exact eigenvector of H_0 with a known energy. It reproduces the four second-order constants, the two-hop weight with the correct sign on three chain types, and the single-contact and shared-link dressings, in seconds — and then gives the corner cumulant to the digit.

Separately, a survey of the corpus located the historical pipeline’s own Stage-3I ledger: 4,221 ordered words whose hash matches the input the kernel copies quote. Its rooted assembly, ported verbatim, reproduces the pinned 189 records exactly. Grouping each word’s rooted images by the set of plaquettes it touches then yields the historical kernel’s own connected cumulants — a reading the original pipeline never performed on its own data.

Theorem 6.3 (Adjudication of the off-axis coefficient). *The historical kernel and the independent assembly agree on every cluster of the rotation record — pair, fourteen chains, two fans, two corners — with identical rationals in both C -parity sectors, except one. The exception is the adjacent-face cube completion. The historical ledger holds 8 of the 24 insertion orderings, each with exactly the weight the multi-loop model assigns it, namely $(2, 1, 4, 2, 8, 8, 2, 4)$ in units of $1/1536$, totalling $31/1536$; the sixteen it lacks sum to $-25/512$. The opposite-face completion in the same ledger has all 24.*

Consequently, in the kernel’s basis,

$$C_{\text{shp}} = C_{\text{historical}} + \frac{25}{1024} = -\frac{13035490122347}{550663802582400} = -0.0236723\dots \quad (6.3)$$

Substituted into the 24 records and solved over the whole zone, this gives $A = 5/48$ and $B = D = 0$ with no residual.

Graph id: C2,ADR:0024 Reproduce: workhouse why C2

[T1 exact], suite “the third implementation and the historical ledger”: nine exact checks, three of them registered findings.

The shortfall has a closed form at every rank. The adjacent-face completion is

$$c_4^{\text{adj}}(N) = \frac{-106}{N(N^2 - 1)^3}, \quad c_4^{\text{adj}}(3) = -\frac{53}{768}, \quad (6.4)$$

in ratio $53/80$ to the opposite-face completion $-160/(N(N^2 - 1)^3)$; the -106 is the primitive -88 plus ten multi-loop orderings at -18 .

6.4 Three things this episode establishes

The disagreement was an enumeration shortfall, not a disagreement about mathematics. A pipeline dropped sixteen of twenty-four orderings on one cluster and nowhere else. That is a diagnosable defect, and it was diagnosed ordering by ordering rather than adjudicated by preference.

A basis error can survive validation. An earlier assembly of the same amplitude was correct and was read in the conjugate basis, giving a third value. Every C -odd component of the historical ledger is the negative of the assembly’s value in one traversal while every C -even one is identical — the signature of conjugating a single face. The assembly was right; its basis was not. The decision record amends it rather than withdrawing it.

The right answer arrived by a forbidden route first. The value (6.3) is the one an all-rank continuation formula produces at $N = 3$ — and that derivation was retracted, because the substitution it used is not permitted. The retraction stands. That the forbidden route happened to reach the number the legitimate assembly later confirmed is not a vindication of the route, and the register keeps both facts.

Scope 6.4. Both recorded kernels remain in the registry with their verdicts: the historical superseded, the cold dump falsified, beside the proven assembled value. Nothing was deleted. One follow-up remains untried — the pair cluster from the third engine, pure-six family included — which can sharpen this result but not overturn it.

7 Rank grading: cancellation and determinant families

The strong-coupling coefficients of §5 and §6 are exact rational functions of N . That makes a question available which is invisible at any single rank: how does each order of the series scale with

the rank, and why?

The answer has two halves. At even orders the scaling is fixed by an exact cancellation among four representation channels. At odd orders the coefficients vanish outright for even N , and for odd N they first appear at order $N - 2$ with a closed-form vertex. Neither statement can be seen from numerics at $N = 3$.

7.1 The one-face grading law

Let $S_m(N)$ be the C -parity splitting of the one-plaquette strong-coupling diagonal at order m ,

$$S_m(N) = (\text{odd})_m - (\text{even})_m = -2 K_m[0][1]. \quad (7.1)$$

Theorem 7.1 (Single-face planar grading through tenth order). *At even orders $m = 2, 4, 6, 8, 10$ the exact rational function of N satisfies*

$$S_m(N) = c_m N^{-(2m-1)}(1 + O(N^{-2})), \quad (7.2)$$

with

$$c_2 = -4, \quad c_4 = -32, \quad c_6 = -\frac{1748}{3}, \quad c_8 = -\frac{123332}{9}, \quad c_{10} = -\frac{49593808}{135}. \quad (7.3)$$

Writing $u = N^2\tau/2$ as in (3.4), the leading part of the order- m contribution is $(c_m/2^m) N\tau^m$. Every even order therefore carries the same single power of N , with

$$b_2 = -1, \quad b_4 = -2, \quad b_6 = -\frac{437}{48}, \quad b_8 = -\frac{30833}{576}, \quad b_{10} = -\frac{3099613}{8640}, \quad b_m = c_m/2^m. \quad (7.4)$$

Graph id: RESULT:SINGLE_FACE_PLANAR_GRADING

Recorded **proven** with **output-certified** evidence. The exponent and the coefficients are exact limits of exact rational functions of N , not numerical extrapolations. An independent integer-rank engine reproduces c_2, c_4, c_6, c_8 by extrapolation to within 1.4×10^{-5} relative — a cross-check by a different method, not the source of the values.

The word *leading* in Theorem 7.1 is doing work and should not be dropped. Each S_m is an exact rational function of N with relative corrections in N^{-2} , so the reduction to τ is a planar-limit statement, not an identity at finite rank. Explicitly, at second order

$$S_2(N) = -4N^{-3} - 4N^{-7}, \quad \text{so} \quad \frac{S_2 u^2}{N} = -\tau^2 - \frac{\tau^2}{N^4}. \quad (7.5)$$

The single-face band per unit N is therefore a function of τ alone as $N \rightarrow \infty$ at fixed τ , and not at finite N . A statement of the second form would be false, and an earlier internal summary of this result made it.

7.2 The grading is a cancellation theorem

Equation (7.2) is a much smaller number than its constituents, and the reason is exact.

Theorem 7.2 (Four-channel cancellation). *Decomposing S_m by the intermediate state of the perturbed vector gives exactly four contributing channels at every even m from 4 to 10: the singlet $((), ())$, the adjoint $((1), (1))$, and the two-box states $((2), ())$ and $((1, 1), ())$. Each is of order $N^{-(m-1)}$, with leading coefficients*

$$(-4, +2, +1, +1) \times 2^{m/2-2}, \quad (7.6)$$

which sum to zero. At $m = 2$ only the singlet and the adjoint contribute, at -2 and $+2$, and these also cancel.

Graph id: RESULT:SINGLE_FACE_PLANAR_GRADING

So the individual channels are of order $N^{-(m-1)}$ and their leading behaviour annihilates, leaving (7.2) at $N^{-(2m-1)}$ — two further orders down in the N^{-2} expansion. At $m = 4$ this is the statement that channels of order N^{-3} conspire to produce a band of order N^{-7} .

This is why the grading is stated as a theorem about a mechanism rather than as an observed power. A power law read off from fitted data at several ranks would carry no information about *why* the leading terms are absent, and would give no reason to expect (7.2) to persist. The cancellation does.

Scope 7.3. Theorem 7.1 concerns the *one-plaquette* diagonal splitting at even orders $m \leq 10$. It is not the cluster cumulant of the multi-face conjecture, which is not discharged by it: the one-face case at sixth order was already established separately, and the shared-link pair route carries the multi-face statement. The general- m law remains a conjecture. All statements are about coefficients at fixed order, not about convergence.

7.3 Odd orders are determinant families

Theorem 7.4 (Centre parity). *For even N , every odd-order matrix element of the des Cloizeaux effective operator of the strong-coupling band vanishes identically — on every cluster, in both C -parity sectors, at every order.*

For odd N , an odd order m vanishes unless $m \geq N - 2$. At $m = N - 2$ the only surviving element is the one-plaquette vertex

$$H_{N-2}(\bar{F}, F) = - \frac{(N/(N+1))^{N-3}}{((N-3)!)^2}, \quad (7.7)$$

which is -1 at $N = 3$ and $-\frac{25}{144}$ at $N = 5$. Hops and leakages vanish at that order.

Graph id: RESULT:ODD_ORDER_CENTRE_PARITY

The mechanism is the centre. The $SU(N)$ Haar integral of a word vanishes unless every link's net flux is $0 \bmod N$; on a single plaquette the four links are private, so the intermediate states are characters with energy $2C_2(R)$, and the flux condition at odd order can be met only once enough boxes have accumulated to close modulo N . That threshold is $N - 2$, and it is why the first odd order is a function of the rank rather than a fixed number.

Scope 7.5. Statements about coefficients of the strong-coupling series at fixed order, not about convergence. On a periodic lattice the link-set lemma requires even extents. The even orders and the overlap theorem are not addressed here.

7.4 Why this refuted a claim

The register records a claim that centre-only circuits are dynamically dark — that configurations carrying only centre flux cannot contribute to the dynamics. Theorem 7.4 and its fifth-order companion refute it: the exact fifth-order determinant correction $\delta c_{5,\det}$ is nonzero. The claim was falsified by an exact computation rather than by an estimate, which is the only way a claim of that shape can be settled.

8 The symbolic-rank engine

Every closed form in §5–§7 is a rational function of N computed as such, not interpolated through ranks. This section explains how, because the method is the most transferable thing in the program and because getting it wrong is the default.

8.1 The problem with sweeping ranks

The natural procedure is to run an exact engine at $N = 3, 4, 5, \dots$ and fit a rational function. That is what was done first. An assembled cluster cumulant β_N agreed with a candidate closed form $P_{17}(N^2)/(N R_{20}(N^2))$ at every rank from 4 to 70: sixty-seven exact agreements between exact rationals.

Sixty-seven exact agreements do not prove the identity. Rational interpolation is unique only once the degrees are bounded, and without a degree bound the agreements constrain nothing about $N = 71$. The review of that work made exactly this objection — no “for all N ” without a degree bound — and it was correct.

A degree bound could have been argued: Weingarten denominators, energy denominators, and a count of vertices together bound the degree, and the sixty-seven agreements would then prove the identity by uniqueness. That argument was not made. What was done instead makes it unnecessary, and is better.

8.2 Running the engine over $\mathbb{Q}(N)$

Inspect where the rank actually enters the exact engine. It enters in exactly two places that are not rational arithmetic:

- the charge predicate, $\text{charge} \equiv 0 \pmod{N}$;
- the flux-difference predicate, $|\text{diff}| \leq N$.

Everywhere else the rank appears only through rational expressions in $C_F = (N^2 - 1)/(2N)$, in $1/(2N)$, and in N^{closed} .

So make the two predicates overridable and promote everything else to the rational function field:

- scalars become elements of $\mathbb{Q}(N)$, carried as a numerator and a monic denominator in exact polynomial arithmetic;
- the Weingarten function becomes the symbolic inverse of the Gram matrix N^{cycles} on S_n ;
- the single-link spectra become the $\text{SU}(N)$ quadratic Casimirs of the irreducible representations that link’s flux content can carry.

A cumulant then emerges as *one rational function of N* , computed rather than fitted. The per-rank engine is otherwise untouched and its own test suite still holds — which is the check that the symbolic run is the same calculation and not a different one.

Theorem 8.1 (Closed forms over $\mathbb{Q}(N)$). *The fourth-order cluster cumulants, the two-hop weight in both C -parity sectors, and the assembled β_N are exact elements of $\mathbb{Q}(N)$ produced by the symbolic engine. The identity $\beta_N = P_{17}(N^2)/(N R_{20}(N^2))$ holds as an identity of rational functions.*

The final polynomial identity is promoted to [T0 Lean], and it is worth saying exactly which identity that is. The corpus *states* $\beta_N = P_{17}(N^2)/(NR_{20}(N^2))$ and never derives it. What Lean checks is that the closed forms the symbolic engine returns for the three cumulants and the adjacent-face cube completion *sum to* the corpus's coefficient — a polynomial identity between the engine's output and the corpus's formula. It is not a Lean derivation of the cumulants, and the Lean source says so in its own docstring. This document repeats the qualification because an internal summary of the same fact dropped it.

Every eigencomponent of every resolvent is verified to be an exact eigenvector of every single-link H_0 , symbolically — a check that has no per-rank analogue, because at a fixed rank a coincidence of numbers is indistinguishable from an identity.

8.3 Why this is the right shape of argument

The methodological content generalizes beyond this program.

An identity between rational functions can be established in two ways: verify it at enough points and bound the degrees, or compute in the function field. The first requires an argument about the size of the objects, which is a second piece of mathematics and is where the difficulty hides. The second requires only that the algorithm's dependence on the parameter be rational — which is a property one can *inspect*, predicate by predicate, rather than estimate.

The title of the decision record is the summary: the degree bound is proved by removing it.

Scope 8.2. The symbolic engine is valid at ranks where the generic partition labelling applies. Exceptional low ranks, where representations degenerate, are handled separately and are the reason $SU(2)$ and in places $SU(4)$ carry their own entries in the register.

This degree bound must not be confused with the *other* degree bound in this program's history — the proposed kinematic mechanism for the fourth-order tier collapse, which was retracted (§6, §15). That one was withdrawn because it was wrong. This one was removed because it became unnecessary.

9 Sixth order: a shape that cannot be removed

The fourth-order shape data closes in a five-element span. The question this section answers is whether the sixth order stays inside it. It does not, and the proof is a divisibility argument that needs no knowledge of the sixth-order dynamics at all.

9.1 Conventions

In the Bloch variables z_j write

$$a_j = 2 - z_j - z_j^{-1}, \quad q = \sum_j a_j, \quad e_2 = \sum_{i < j} a_i a_j, \quad e_3 = a_1 a_2 a_3, \quad (9.1)$$

and let $U = \psi\psi^*$ with $\psi^*\psi = q$, $P_c = U/q$, $Q_c = I - P_c$, $S = L_\downarrow - 4I$, and R the cross-plane part of S .

Two warnings, both of which have caused errors here. The invariants e_2, e_3 are Bloch invariants, *not* representation Casimirs; and U is the up-Laplacian, not its normalized projector, so its kernel is the complementary plane rather than the carrier line. Cleared identities hold at $q = 0$; normalized carrier formulas require $q > 0$.

9.2 The complete folded operator

Let P_E be the isolated electric model-space projector, $D = Q_E/(E_0 - H_0)$, and V_0 the perturbation. If $P_E V_0 P_E = a P_E$, set $V = V_0 - aI$ on the whole Hilbert space: shifting the complementary block as well is essential. With $\chi_0 = P_E$, intermediate normalization gives the exact recursions

$$K_n = P_E V \chi_{n-1}, \quad \chi_n = D V \chi_{n-1} - D \sum_{j=1}^{n-1} \chi_{n-j} K_j, \quad (9.2)$$

and with $M = P_E + \chi^* \chi$ the canonical Hermitian effective operator is $M^{1/2} K M^{-1/2}$. Writing

$$A_m = P_E V (D V)^{m-1} P_E, \quad C_2 = P_E V D^3 V P_E, \quad (9.3)$$

and B_m for the sum of A_m with exactly one of its $m - 1$ slots replaced by D^2 :

Theorem 9.1 (The eighteen-word sixth-order fold).

$$H_6 = A_6 - \frac{1}{2}\{A_4, B_2\} - \frac{1}{2}\{A_3, B_3\} - \frac{1}{2}\{A_2, B_4\} + \frac{1}{2}\{A_2^2, C_2\} + \frac{3}{8}\{A_2, B_2^2\} + \frac{1}{4}B_2 A_2 B_2. \quad (9.4)$$

Expanded, this is exactly eighteen words in P_E, V, D . The same recursion reproduces the established $H_4 = A_4 - \frac{1}{2}\{A_2, B_2\}$.

Graph id: RESULT:G9_HERMITIAN_FOLDED_SUPPORT

The $\{A_3, B_3\}$ term is present and it matters: dropping odd electric histories gives an incomplete formula. No odd order is dropped anywhere in the recursion. Independent rational matrix checks verify both the invariant-subspace equation and the normalization through degree six, with a rank-two P_E and non-commuting effective coefficients.

Does not assert. Equation (9.4) is the universal folded operator support in this convention. It does not evaluate Wilson plaquette matrix elements, establish a specific support, or carry out connected multi-plaquette rooted subtraction. Those require the actual electric and Haar implementation on each support.

9.3 Exact word symbols

In a polynomial carrier gauge, exact symmetrization gives each word its cleared symbol in $\mathbb{Q}[q, e_2, e_3]$. The table below is computed, not inferred, and all forty words of length zero through three agree coefficient by coefficient with independently implemented spatial Laurent operators.

Word	$\sigma(W)$	Word	$\sigma(W)$
I	q	RR	$q e_2 + 3 e_3$
U	q^2	RUR	$4 e_2^2$
S	$-4q$	RSR	$(q - 4)(q e_2 + 3 e_3) - 4 e_2^2$
S^2	$16q$	RRR	$-2 e_2^2 - 2 q e_3$
R	$-2 e_2$	URR, RRU	$q^2 e_2 + 3 q e_3$
UR, RU	$-2 q e_2$		

A word census must retain coefficients and combine them before testing shape membership: $\sigma(RUR + 2 RRR) = -4 q e_3$, although both individual words contain e_2^2 . Support is not unique modulo operator identities, and the invariant object being tested is the complete carrier polynomial. Missing direct input is represented as unknown, not as zero.

9.4 The band coefficient at sixth order

At fixed nonzero momentum, with scalar anchors removed,

$$H_{\text{eff}}(u) = u^2 t_3 q Q_c + u^3 H_3 + u^4 H_4 + u^5 H_5 + u^6 H_6 + O(u^7). \quad (9.5)$$

The recorded $\text{SU}(3)$ third-order factorization gives $Q_c H_3 P_c = 0$, so the off-diagonal block first appears at order u^2 in the scaled operator, from H_4 . Expanding the Schur complement for the scalar carrier branch,

$$\lambda_6 = \frac{\sigma(H_6)}{q} - \frac{q \sigma(H_4^2) - \sigma(H_4)^2}{t_3 q^3}. \quad (9.6)$$

This is not an assumption that H_5 vanishes. H_5 first couples to H_4 at order u^7 in H_{eff} , as does the uH_3 correction to the complementary inverse, and unknown off-diagonal H_6 contributes later still. The decoupling of H_3 is an explicit recorded input rather than an assumption, and any scalar vacuum subtraction is part of H_6 and changes nothing.

Evaluating all twenty-five ordered pairs from the actual H_4 support $\{I, U, S, S^2, R\}$, including their projected subtraction, gives zero for twenty-four of them. Only RR survives, in agreement with direct composition of the full assembled spatial kernel. With

$$\kappa = \frac{4C_{\text{shp}}^2}{t_3} = \frac{169924002729806205028788409}{619343593697933825385600000} > 0, \quad t_3 = \frac{5}{612}, \quad (9.7)$$

the known dynamical mixing support is $-(\kappa/q)RR + (\kappa/q^2)RUR$, with normalized expectation

$$\lambda_6^{\text{mix}} = -\kappa \left(\frac{e_2}{q} + \frac{3e_3}{q^2} - \frac{4e_2^2}{q^3} \right). \quad (9.8)$$

Does not assert. The RUR here arises from inserting $P_c = U/q$ in the projected subtraction. It is not evidence that a direct six-insertion electric history carries the local word RUR , and the ratio $1 : 3 : -4$ belongs to this mixing term rather than to the full coefficient.

9.5 Non-cancellation

Theorem 9.2 (An extra rational shape survives every local direct term). *Assume the order-six electric-shell effective operator is a finite-range, translation-covariant stencil with Laurent-polynomial entries in this basis, and write $F = \sigma(H_6)$. Then (9.6) gives*

$$q^3 \lambda_6 = q^2 F - \kappa(q^2 e_2 + 3q e_3 - 4e_2^2), \quad (9.9)$$

and q does not divide the right-hand side, for any Laurent polynomial F .

Proof. Modulo q , the right-hand side of (9.9) is $4\kappa e_2^2$. Evaluate at

$$z_1 = z_2 = -1, \quad z_3 = 5 + 2\sqrt{6}, \quad (9.10)$$

all nonzero, so the point lies in the Laurent ring. Since $z_3^{-1} = 5 - 2\sqrt{6}$, we get $a = (4, 4, -8)$, hence $q = 0$ and $e_2 = -48$. The value there is $4\kappa \cdot 48^2 = 9216\kappa \neq 0$, and a Laurent polynomial divisible by q would vanish at that point. $\square$

Graph id: RESULT:G9_LOCAL_SIXTH_NONCANCELLATION

[T1 exact]. Consequently λ_6 cannot lie in the fourth-order shape span $\{1, q, e_2, 4e_2/q, e_3/q\}$, and no local direct RUR term removes the extra rational shape: a direct $hRUR$ contributes $4he_2^2/q$, which cannot cancel an e_2^2/q^3 component. For symmetric F the remainder modulo q^2 is exactly $4\kappa e_2^2 - 3\kappa q e_3$, so both e_3/q^2 and e_2^2/q^3 are present. The e_2/q term alone *can* be cancelled, by $F = \kappa e_2$ — for instance a local $H_6 = -(\kappa/2)R$ — so the $1 : 3 : -4$ ratio is not a constraint on all sixth-order shapes.

Note what the proof does not use: any knowledge of F . The extra shape is forced by the fourth-order data alone, through the Schur complement, whatever the sixth-order dynamics turns out to be.

9.6 A withdrawn derivation

An earlier version of this work enumerated closed rooted walks — 90 four-step unit-edge walks, of which 66 retrace immediately and 24 are planar squares, together with 192 non-retracing six-step walks using all three axes — and read a sixth-order amplitude off that census.

The derivation is withdrawn. Edge walks are not ordered plaquette insertions with electric states, Haar weights, projectors and folds, and no map between them was supplied; the census therefore determines no net coefficient of any of the candidate words. The old amplitude formula was assigned rather than derived. The old carrier function used $\sigma(R) = 0$ and $\sigma(U) = q$, where the actual unnormalized symbols are $-2e_2$ and q^2 ; it returned a nonzero expression for a defect it separately reported as zero. The implementation now computes the symbols and tests the defect, and the old interface raises an explicit missing-dynamics error rather than returning a number.

What is withdrawn is a claimed derivation, not the possibility of a physical amplitude of that shape. Theorem 9.2 is independent of the census entirely. Computing the direct term on a connected multi-plane support is Problem 8, and it is the cheapest decisive computation currently open in this program.

10 A rank-uniform local shell construction

This section is a separate result in its own regime, included because it is exact, rank-uniform, and independent of everything above — not because it continues it. The object is the formal local Wilson oscillator expansion around a single plaquette, and the point is that its spectrum admits a finite exact recursion at every perturbative order with coefficients in $\mathbb{Q}[N, N^{-1}]$, needing no Gram-matrix inversion.

Conflating this construction with the flux band of §4–§9, or with the Wilson transfer operator of §12, would be an error. They are four different objects — the physical $SU(N)$ transfer band, the $SU(3)$ internal Hamiltonian carrier, the conditional block, and this local oscillator — and no cross-regime continuation between them is asserted anywhere in this document.

10.1 The exact algebra

Let X be a traceless Hermitian $N \times N$ matrix with density proportional to $e^{-\text{Tr}(X^2)}$, write $P_k = \text{Tr}(X^k)$ with $P_0 = N$ and $P_1 = 0$, and use a traceless Hermitian basis normalized by $\text{Tr}(T_a T_b) = \delta_{ab}$. The covariance is

$$\mathbb{E}[X_{ab} X_{cd}] = \frac{1}{2} \left(\delta_{ad} \delta_{bc} - \frac{1}{N} \delta_{ab} \delta_{cd} \right). \quad (10.1)$$

With $D = \frac{1}{2}\Delta$ and E the polynomial Euler operator, the completeness identity for this basis gives the exact relations

$$DP_k = \frac{k}{2} \sum_{l=0}^{k-2} P_l P_{k-2-l} - \frac{k(k-1)}{2N} P_{k-2}, \quad (10.2)$$

$$D(fg) = (Df)g + f(Dg) + \nabla f \cdot \nabla g, \quad (10.3)$$

$$\nabla P_k \cdot \nabla P_l = kl \left(P_{k+l-2} - \frac{1}{N} P_{k-1} P_{l-1} \right). \quad (10.4)$$

These are exact at every integer rank, including ranks at which some traces become dependent — which is the property that makes the construction rank-uniform rather than rank-by-rank. For homogeneous F of positive degree d , Gaussian integration by parts gives $\mathbb{E}_\mu[F] = \mathbb{E}_\mu[DF]/d$, and with $\mathbb{E}_\mu[1] = 1$ and vanishing odd moments this is a finite exact moment recursion.

10.2 Shell projectors without a Gram inverse

Proposition 10.1 (Terminating Wick conjugation). *Put $L = E - D$. Since $[E, D] = -2D$, the terminating exponential $W = \exp(-D/2)$ satisfies*

$$LW = WE. \quad (10.5)$$

Hence, with H_d the extraction of the homogeneous degree- d part,

$$\Pi_d = \exp(-D/2) H_d \exp(D/2) \quad (10.6)$$

is the exact oscillator-shell projector, and on each finite polynomial space $R_s = \sum_{d \neq s} \Pi_d / (s - d)$ is the exact shell resolvent.

The content of Proposition 10.1 is that L and E are conjugate by a *terminating* exponential, so the shell decomposition of the interacting operator is the shell decomposition of the Euler operator, transported. No Gram matrix is formed and none is inverted, which is precisely the step that makes rank-uniformity delicate elsewhere in this program.

10.3 The coefficients

Theorem 10.2 (All-rank local coefficients). *The terminating conjugation gives exact shell projectors and a unique all-order simple-branch formal recursion in $\mathbb{Q}[N, N^{-1}]$, with state degree at most $s + 4r$. The two formal local gaps have exact all-rank coefficients through β^{-2} , including c_3^- and $c_4^\pm$. The five corrections $c_0, \dots, c_4$ are strictly negative at every allowed rank, by positive shifted numerator polynomials.*

Graph id: RESULT:LOCAL_CLASS_WICK_RECURRENCE, RESULT:LOCAL_CLASS_WICK_COEFFICIENTS, RESULT:LOCAL_CLASSES_WICK_SIGNS

The construction applies to the vacuum, the first charge-even shell for $N \geq 2$, and the first charge-odd shell for $N \geq 3$. It recovers the previously archived c_0, c_1, c_2 formulas in both sectors and the archived all-rank c_3^+ , and additionally derives c_3^- and both all-rank c_4 formulas. The certificate carries full polynomial eigen-equation residuals rather than only agreement with the archived constants — a distinction worth stating, because agreement with a previous value is a weaker check than satisfying the equation.

The recovery is by an independent route: the archive engine uses full-GUE cut-and-join and then subtracts the trace, while this one applies D directly in traceless coordinates.

10.4 A companion algebraic identity

Theorem 10.3 (Shifted-power carrier symbols). *For the cubic Laurent operators with $S = L_\downarrow - 4I$,*

$$\sigma(RS^m R) = (q - 4)^m (qe_2 + 3e_3) - 4\Pi_m(q) e_2^2 \quad (10.7)$$

for every natural m , with Π defined polynomially also at $q = 0$.

Graph id: RESULT:RSM_R_CARRIER_SYMBOL_MASTER_THEOREM

This closes the word table of §9 at all powers, and is formalized in Lean as an all-power operator statement.

Scope 10.4. Theorem 10.2 concerns the *formal* local class oscillator only. No error bound for the compact-group eigenvalues is claimed, and local spectral coefficients do not by themselves identify the interacting Wilson spectrum, an interacting-volume estimate, or a continuum source measure. Theorem 10.3 is an algebra identity and carries no sixth-order dynamical population claim.

11 External comparison

Published work enters this program as evidence rather than as authority. A paper is unchecked until something checks it, exactly like an internal document; the reason an external agreement counts for more is not that it is published but that it was produced without knowledge of this program, so agreement is not bookkeeping.

The map holds 108 indexed papers and 152 recorded relations to named claims here. 54 of those edges are verified — the paper was obtained, read, and compared against the claim — across 34 distinct papers. The remainder are recorded targets for papers not yet obtained, which is itself useful: a paper heavily cited within this web and still unobtained is automatically the next acquisition target.

An entry earns its place by naming a specific claim and saying what relationship it has to it. `workhouse lit -for G19` answers the query in the other direction, and validation rejects an entry whose target does not resolve.

11.1 A 1989 table, and rationals computed cold

The sharpest external check available to this program is a Hamiltonian strong-coupling series for SU(3) glueball masses published in 1989 [51], whose Table 1 is pinned here by digest. Under the normalization bridge $m_n = 2^{n-1}a_n$, its coefficients are compared against the Γ -point coefficients this program derives as exact rationals.

In the C -odd (1^{+-}) channel — the sector of the carrier, whose series is (4.5) — the agreement runs through third order:

Order	Published	This program	Gap (bound)
$n = 0$	$\frac{16}{3}/2$	$\frac{8}{3} = E_{\text{flat}}(0)$	exact
$n = 1$	$a_1 = +1$	the u coefficient +1	exact
$n = 2$	$2a_2 = 0.0359477124184$	$\frac{11}{306}$	$9.94 \times 10^{-14} \ (2 \times 10^{-13})$
$n = 3$	$4a_3 = -0.437135556836$	$-\frac{109151}{249696}$	$1.11 \times 10^{-12} \ (3 \times 10^{-12})$

In the C -even (0^{++}) channel, $2a_2 = -0.21274509804$ against $-\frac{217}{1020}$ with gap 7.84×10^{-13} , and $4a_3 = -0.1039004229144$ against $-\frac{54049}{520200}$ with gap 1.36×10^{-13} . The sign structure matches as well: $a_1 = -1$ in the scalar channel, where the carrier's is $+1$.

Each bound above is the printed half-ulp of the published table scaled with margin, so these are agreements to the precision at which the 1989 table was printed — not approximate corroborations. The rationals $11/306$ and $109151/249696$ are the third-order coefficients of E_{flat} in (4.5), computed here from Haar integration and the cellular complex with no knowledge of the table.

At fourth order, $8a_4 = -0.7751458630184$ against $m_{\Gamma}^{(4)} = -0.7751458630189173$, gap 5.17×10^{-13} against a bound of 5.3×10^{-13} .

Scope 11.1. These are [T2 numeric] checks: float agreement within a stated tolerance, to the precision of a printed table. They corroborate the coefficients; they do not prove them, and the exact statements of §5–§7 do not rest on them. Note also that the pinned revision of the source compares only $2a_2$ and $4a_3$ in its own text — the fourth-order agreement is registered here and not leaned on there.

11.2 The two contradicting edges

Two edges in the map are recorded as *contradicts*, both from a 1976 strong-coupling series, and both against this program: one against the third-order C -odd coefficient, one against the Γ -point fourth-order value. They originate in a disagreement the 1989 paper itself reports at orders x^3 and x^4 with the series it supersedes.

Both are recorded as not yet obtained. Until that primary source is read, this program has two unresolved external contradictions against coefficients it otherwise regards as settled, and says so rather than resolving them by the age of the sources. Obtaining that paper is listed with the acquisition targets below.

11.3 How far a novelty claim here can be trusted

Not as far as the size of the index suggests, and the limitation is worth stating precisely rather than as a disclaimer.

Only 54 of 152 recorded edges are verified. The unverified remainder is not a random sample: it includes several of the *nearest* neighbours to this program's own results. The main Nuclear Physics B paper of the 1981 strong-coupling glueball series is recorded as not obtained — only its errata were — and the 1974 and 1979 large- N papers that bear most directly on the rank grading of §7 are likewise unobtained, with the ledger noting for one of them that its definitions were used from memory and only their sign consequences checked.

Consequently: where this document says a result is novel, that means *no verified entry in this index contains it*, which is a weaker statement than priority. Two of the strongest results are affected. The rank-grading cancellation of §7 sits closest to large- N counting arguments in papers this program has not read; and the fixed-spacing transfer construction of §12 sits alongside the classical positivity and self-adjoint transfer-matrix results of Osterwalder and Seiler, which the index does register.

Acquiring the unobtained near neighbours is cheap relative to everything else in §16, and it is the one item on that list that could *subtract* from the program rather than add to it. That is a reason to do it, not a reason to defer it.

Table 1: Verified external-evidence edges. Each row is a published result that was obtained, read, and checked against a named claim of this program. Relation is the ledger’s own classification; *contradicts* and *confusable* are listed first because they are the rows that cost the program something.

Paper	Ref.	Year	Bears on	Relation
HSB_2000	[53]	2000	HAMER_A4_NUM	confusable with
HAMER_1989	[51]	1989	BAND_EVEN_BOTTOM	corroborates
HAMER_1989	[51]	1989	C1	corroborates
LLL_2006	[69]	2006	C1	corroborates
HAMER_1989	[51]	1989	D_3	corroborates
SCHIERHOLZ_1988	[91]	1988	G18	corroborates
HAMER_1989	[51]	1989	M3_EVEN_K0	corroborates
KPS_1981	[64]	1981	R14	corroborates
CBB_2026	[21]	2026	R2	corroborates
KPS_1981	[64]	1981	G7	supplies a value
HAMER_1989	[51]	1989	HAMER_A4_NUM	supplies a value
MUNSTER_1985_TM	[79]	1985	C2	supplies a comparison
CAO_2023	[16]	2023	C7	supplies a comparison
LEMOINE_2026	[68]	2026	C7	supplies a comparison
HAZRA_2026	[54]	2026	G14	supplies a comparison
AT_2020_SU3	[2]	2020	G18	supplies a comparison
BESIII_2026	[22]	2026	G18	supplies a comparison
CHEN_2006	[18]	2006	G18	supplies a comparison
LLL_2006	[69]	2006	G18	supplies a comparison
MORNINGSTAR_2025	[76]	2025	G18	supplies a comparison
MP_1999	[77]	1999	G18	supplies a comparison
AT_2021_SUN	[3]	2021	G19	supplies a comparison
BB_1983	[10]	1983	G19	supplies a comparison
FLPS_2026	[27]	2026	G19	supplies a comparison
URZUA_2019	[102]	2019	G19	supplies a comparison
BORGA_2024	[11]	2024	G5	supplies a comparison
CAO_2023	[16]	2023	G5	supplies a comparison
OBZ_1985	[82]	1985	G5	supplies a comparison
AT_2021_SUN	[3]	2021	G6	supplies a comparison
CM_2003	[17]	2003	G6	supplies a comparison
BALAJI_2026	[8]	2026	R2	supplies a comparison
CKS_2021	[19]	2021	U1	supplies a comparison
DRS_2021	[25]	2021	U1	supplies a comparison
KRS_2023	[62]	2023	U1	supplies a comparison
SZH_1997	[92]	1997	U1	supplies a comparison
CS_2006	[23]	2006	C7	supplies a method
JW_2006	[61]	2006	G17	supplies a method
UELTSCI_2004_CLUSTER	[101]	2004	G17	supplies a method
YAROTSKY_2004_GS	[107]	2004	G17	supplies a method
BB_1983	[10]	1983	G18	supplies a method

Paper	Ref.	Year	Bears on	Relation
JW_2006	[61]	2006	G18	supplies a method
NSY_2019	[80]	2019	G18	supplies a method
YAROTSKY_2004_QP	[108]	2004	G18	supplies a method
JW_2006	[61]	2006	G19	supplies a method
KLEIN_ROSENBERGER_2020_WKB	[63]	2020	G19	supplies a method
SCHIERHOLZ_1988	[91]	1988	G19	supplies a method
UELTSCHI_2004_CLUSTER	[101]	2004	G19	supplies a method
YAROTSKY_2004_GS	[107]	2004	G19	supplies a method
YAROTSKY_2004_QP	[108]	2004	G19	supplies a method
JW_2006	[61]	2006	G20	supplies a method
JW_2006	[61]	2006	G22	supplies a method
JW_2006	[61]	2006	G23	supplies a method
MUNSTER_1985_TM	[79]	1985	G3	supplies a method
KRS_2023	[62]	2023	R2	supplies a method

Part II

The construction at fixed lattice spacing

12 The construction at fixed lattice spacing

This is the strongest constructive statement in the program. It is also the one most often misread, so it is stated here with its hypotheses, its interval, and an explicit list of what it does not assert.

Throughout this section ε denotes the temporal mesh. It is *not* the planar variable τ of (3.4); the two are unrelated and the symbols are kept distinct deliberately.

12.1 The theorem

Theorem 12.1 (Infinite-volume Wilson band and literal source frame). *Under the hypotheses listed below, the Perron-normalized Wilson transfer operator has a common strong limit along open-box and centered-periodic exhaustions, in the transported product representation. The limit is a positive contraction fixing the vacuum; it differs from the free product in norm by at most $1/998$, and its norm on the non-vacuum subspace is at most $\frac{4}{5} + \frac{1}{998}$.*

On the coupling interval

$$|u| \leq \frac{u_*}{10\,022\,400\,000\,N}, \quad (12.1)$$

the complete infinite-volume physical odd band is isomorphic to $\ell^2(\mathbb{Z}^3) \otimes \mathbb{C}^3$. The projected literal plaquette-source synthesis is onto that entire band, with Gram bounds

$$\frac{9}{16} I < \mathcal{G} \leq \frac{81}{64} I. \quad (12.2)$$

The Euclidean Osterwalder–Schrader band and the quantum GNS band coincide, intertwining the physical transfer and the sources.

Graph id: RESULT:WILSON_INFINITY_PHYSICAL_BAND

Recorded **proven** with **analytic** evidence; [T3 analytic] for whole-statement machine certification, with its component estimates checked across several suites.

12.2 Hypotheses

1. All calibrated kinetic, positive Wilson-kernel, incidence and common-disk hypotheses of the weighted-activity and cardinality-chart theorems.
2. Fixed $SU(N)$ with $N \geq 3$, and fixed *spatial* lattice spacing. Compatible induced open boxes, or centered periodic exhaustions, carrying the same local data.
3. Temporal mesh $0 < \varepsilon \leq \varepsilon_0 \leq \log(5/4)/(4\gamma)$, with $m = \lceil \log(5/4)/(\gamma\varepsilon) \rceil$, and the full neutral physical odd free-window premise.
4. $\gamma = C_F/2 = (N^2-1)/(4N)$, the *exact* calibrated fundamental-link energy. This is a hypothesis about an identity, not a gap lower bound one is free to weaken: substituting a smaller number invalidates the conclusion rather than weakening it.
5. Literal odd sources $\mathcal{O}_p = (\chi_p - \bar{\chi}_p)/\sqrt{2}$, with free orthonormal plaquette vectors of norm at most $\sqrt{2}N$.
6. The calibrated representation-uniform kinetic-window mesh premise, whose N -dependent admissible temporal threshold is the previously established one. Every estimate is uniform in volume and in meshes satisfying these premises, and the common coupling interval depends on fixed N and is nonzero.

The interval (12.1) unpacks as $u_0 = u_*/1252800000$ and $u_1 = u_0/(8N)$, with the accompanying constants $\eta_{\text{act}} = 1/2500$, $\epsilon = 1/998$, $g_* = 1024/15625$ and $q_0 = \frac{4}{5} + \frac{1}{998} < 1$. The bound $\eta_{\text{act}} \leq 1/2500$ on the integrated generator norm at weight $1 + \log 2$ is a *hypothesis* of the source specialization, not a conclusion of the theorem; it is listed here to prevent the reading in which the theorem is taken to establish it.

12.3 Why totality is the substance

The clause that carries the physical content of Theorem 12.1 is that the source synthesis is *onto* the band, not merely bounded into it.

Break 14.4 is the reason. A local probe family can decay at one rate while the true spectrum contains a much slower state the probe cannot see; the three-dimensional model of Proposition 14.4 exhibits a factor of twenty-one between the observed rate and the actual gap. A construction that produced a band and a decay estimate, without totality, would prove nothing about the spectrum.

Totality here is a conclusion. The tagged source norm controls the entire synthesis operator; close projections and a Neumann inverse establish surjectivity in infinite dimension; and reflection-positive history completion contains the whole band by selective source cyclicity. Fine time steps are the positive m -th root of the block.

12.4 What Theorem 12.1 does not assert

Does not assert. Global equality of the Osterwalder–Schrader and full quantum high-energy spaces. Global transfer injectivity. Any spatial continuum limit. Extension beyond $SU(N)$, $N \geq 3$. The positive energy gap is in electric-time units, and its conversion to physical units is not part of the statement.

Most importantly, and separately from the list above: the interval (12.1) is a neighbourhood of $u = 0$, and the continuum limit is taken along a trajectory on which $u \rightarrow \infty$. Theorem 12.1 therefore supplies no control whatever on the continuum limit. This is Break 14.5, it is structural, and no improvement of the constant 10 022 400 000 addresses it. A version of exactly that inference was made in this program’s own drafts and retracted; see §15.

12.5 The relation to the Hamiltonian carrier

Theorem 12.1 is a statement about the Wilson transfer operator. The Hamiltonian carrier of Part I is a different object, spectrally identified with the band only on momentum patches away from zero.

The obstruction at zero momentum is proved, not merely encountered, and it is a statement about *chains*:

1. every vector in the image of the cube boundary map has exactly zero zone-average component;
2. the zero-momentum carrier content is therefore carried entirely by the three wrapping sheets of §4;
3. a sheet’s normalized overlap with any fixed local observable is exactly $1/L$;
4. compact two-cycles on open windows are exactly cube boundaries — checked directly for small windows, and Alexander duality in general.

The repository’s own prose summarizes (i)–(iv) as “every finitely supported exactly-flat operator is Γ -blind”. That phrasing is a convenient shorthand and this document does not adopt it: the four statements are about chain vectors, and read literally as a claim about arbitrary operators the shorthand is false. What follows from (i)–(iv) is the thing actually needed — that an interpolator for the zero-momentum protected level must leave the exactly-flat local algebra, coupling either through in-sector directions that are flat-degenerate at Γ anyway, or through out-of-sector multi-plaquette content.

12.6 A prediction, sharper than the measurement it explains

The published few-percent overlap of a local operator with the lightest state is not a constant: it degrades toward the continuum. For a local lattice source whose lowest-dimension continuum operator has dimension d , the overlap law is

$$Z_0(a) \sim \frac{|f_d|^2 a^{2d-3}}{2M}, \quad (12.3)$$

which at $d = 4$ gives a^5 and so reproduces the published measurement — the agreement that makes (12.3) worth recording at all.

But a^5 is the wrong exponent for *this* carrier. The C-odd carrier’s leading continuum operator is $d^{abc} F^a F^b F^c$, of dimension six, so the applicable exponent is $a^{2 \cdot 6 - 3} = a^9$. The extrapolation is brutal: $Z_0 \approx 1.2 \times 10^{-6}$ at $a\sqrt{\sigma} = 0.10$ and 2.4×10^{-9} at 0.05. Still a power, so formally survivable, and a serious practical obstacle to measuring this state with a local operator.

The same law is the argument for smearing, quantitatively: at fixed physical radius R , $Z_0 = |f_R|^2 R^{2d-3}/(2M)$ is a -independent. A smeared basis is structural here, not a convenience.

Scope 12.2. Equation (12.3) is asserted, not proved here. What is checked is the dimension arithmetic and the agreement of its $d = 4$ case with the published a^5 . An earlier version of this program’s notes quoted a^5 for the carrier itself; that is corrected above.

Scope 12.3. The out-of-sector dressing, the transfer-matrix identification at zero momentum, and everything non-perturbative remain open. A particle statement lives at fixed nonzero momentum, where no collapse occurs; the L^{-2} statement in the literature is a minimal-grid-momentum statement governing the approach to Γ , not a fixed-momentum one.

Part III

The continuum limit: what is required, and what is ruled out

13 The continuum problem

Part II constructs an infinite-volume theory with a positive gap at fixed lattice spacing. This part is about the distance between that and a continuum quantum Yang–Mills theory with a mass gap, and it is written to be useful to someone trying to close that distance.

13.1 The target, and what has no argument here

The statement in question is the existence of a non-trivial quantum field theory on $\mathbb{R}^{1,3}$ for $G = \text{SU}(N)$, $N \geq 2$, satisfying the Wightman axioms, with

$$\text{spec}(H)|_{\Omega^\perp} \subset [m, \infty), \quad m > 0. \quad (13.1)$$

Four of its ingredients have no argument anywhere in this program and are stated here so that no reader has to infer their absence.

1. **Poincaré covariance.** No construction of a unitary representation of the covering group of $\mathcal{P}_+^\uparrow$ is given.
2. **Restoration of rotational invariance.** The hypercubic lattice breaks it; nothing here restores it in the limit.
3. **The continuum measure.** It is never constructed.
4. **Reconstruction.** Even granting a limiting kernel, the passage to a Wightman theory is not carried out.

What *is* carried out is a set of estimates on the lattice side, and this part states them, states why the six obstructions of §14 block their assembly, and states in §16 what would have to be proved.

13.2 The three routes that are live

The estimates in this program group into three approaches, which are alternative rather than sequential. The ledger records them as independent routes and explicitly notes that no chain of necessary conditions runs between them.

The conditional-score route. Control the conditional law of a link given its complement, in the small-coupling limit, by dominating the score of the conditional measure. The available inputs

are strong: exact heat-bath conditional laws, synchronized radial dilation following the conditional minimum curve, the antipodal magnetic normal geometry, exponential suppression outside the complete minimizing set, and a uniform Agmon action gap on the complement. The obligation is Problem 1, and what stands between the inputs and the conclusion is polynomial amplitude control and the tube moments, not a missing structural idea.

The transport route. Compare the fine and coarse theories by transporting the vacuum and the sources along a common energy domain against a physical clock. Here the recorded failure is instructive: an earlier closure used a vacuum-only skew generator, and the source generator transports the ground state rather than annihilating it. The corrected obligation is Problem 2, and its companion, Problem 3, is the requirement that the comparison’s constants not degrade with the number of interacting blocks.

The moving-time route. Extract a limiting gap from approximate sources and one growing observation time per cutoff. The implication from its hypotheses to a gap is proved, and sharp: pairwise inequalities give the optimal normalized rate, the horizon penalty and a witnessing time, and one-atom examples show the rate cannot be improved. Its hypotheses are the difficulty — see Break 14.3, which shows the theorem returns exactly zero when the background error rate fails to exceed twice the probe tilt rate, an inequality no calculation certifies for four-dimensional Yang–Mills.

13.3 Why none of the three currently closes

Each route reaches a point where a summability or uniformity statement is required, and at each such point the natural estimate is one power short.

That is not a coincidence of technique. Break 14.1 exhibits the arithmetic: on an asymptotically free trajectory the running coupling gives increments of size $O(g_k^2) = O(1/k)$, the normalization of the scale derivative carries no small factor, and $\sum 1/k$ diverges. Break 14.2 exhibits the same shortfall in the polymer variable, where a contracting activity and a scale-independent physical mass cannot be read off one schedule. Problem 3 is the same shortfall again in the block count.

Underneath all three sits Break 14.5: the estimates are proved near $u = 0$ and the continuum lives at $u \rightarrow \infty$. A summable increment would connect scales *within* a regime; it would still not move the construction between regimes. That is why Problem 6 is marked load-bearing and the others are not.

14 Six obstructions, with counterexamples

This section states the obstructions that any argument from the fixed-spacing construction of §12 to a continuum mass gap must clear. They were produced by a deliberate adversarial audit of this program’s own drafts, and four of them killed arguments that had been written down here as complete.

They are stated as mathematics, not as caveats. Five carry explicit finite counterexamples, and a finite counterexample to a proof strategy is a permanent fact about that strategy. A reader who wants to know where the remaining difficulty of the problem actually sits should read this section and §16 and can safely skip everything else in Part III.

The counterexamples are not rhetorical. Those of Break 14.4 are registered checks in this repository’s own test suite — `tests/test_os_kernel_gap.py`, six checks, passing — and the two collapse conditions of Break 14.3 are checked in `tests/test_moving_time_gap.py`. An obstruction

that a repository asserts against itself, and then runs in continuous integration, is harder to forget than one recorded in prose.

14.1 Break 1: an $O(g_k^2)$ increment bound does not give convergence

A renormalization-group argument for the continuum limit produces a sequence of effective actions S_k^{eff} and observable expectations $m_k(\mathcal{O})$ indexed by scale, and controls the one-step increment by the running coupling:

$$|m_{k+1}(\mathcal{O}) - m_k(\mathcal{O})| \leq C_{\mathcal{O}} g_k^2 \leq \frac{C'_{\mathcal{O}}}{k + k_0}, \quad (14.1)$$

the second inequality being asymptotic freedom, $g_k^2 = \kappa/(k + k_0)$. The step that must not be taken is to conclude from (14.1) that (m_k) is Cauchy. Telescoping gives only

$$|m_n(\mathcal{O}) - m_m(\mathcal{O})| \leq C'_{\mathcal{O}} \sum_{k=m}^{n-1} \frac{1}{k + k_0} \asymp C'_{\mathcal{O}} \log\left(\frac{n}{m}\right), \quad (14.2)$$

which does not tend to zero as $m, n \rightarrow \infty$.

Proposition 14.1 (Counterexample). *Let $x_k = \sin(\log(k + 1))$. Then $x_k \in [-1, 1]$ and, by the mean value theorem, $|x_{k+1} - x_k| \leq 1/(k + 1) \rightarrow 0$, yet the set of subsequential limits of (x_k) is the whole interval $[-1, 1]$.*

So the increment bound (14.1) is consistent with the observable expectations oscillating forever across a fixed interval. What is needed is one more power: $|m_{k+1} - m_k| = O(g_k^4) = O(k^{-2})$.

The obstruction to obtaining it is not technical. On the asymptotically free trajectory $u_k = (\alpha + \beta k)^{-1}$,

$$\Delta u_k = -\frac{\beta}{(\alpha + \beta k)(\alpha + \beta(k + 1))}, \quad \frac{|\Delta u_k|}{4u_k u_{k+1}} = \frac{\beta}{4} = O(1), \quad (14.3)$$

and differentiating an expectation along the trajectory,

$$\frac{d}{ds} \mu_s(\mathcal{O}_s) = \mu_s(\partial_s \mathcal{O}_s) - \text{Cov}_{\mu_s}(\mathcal{O}_s, \partial_s S_s), \quad (14.4)$$

the normalization carries no small factor. The natural size of the variation is therefore $O(g_k^2)$, and a bounded-covariance estimate cannot improve it. Converting $1/k$ into $1/k^2$ requires an exact cancellation of the leading partition-function covariance in (14.4). No such cancellation is proved here.

Scope 14.2. This obstruction applies to the multiscale route recorded as

DERIV:YANGMILLS_CONTINUUM_BALABAN_MULTISCALE_PROOF:THEOREM_7_1,

whose Cauchy input is exactly what (14.1) fails to supply. It does not bear on any statement in Part I or II, which are at fixed scale.

14.2 Break 2: the polymer activity and the mass schedule are algebraically incompatible

A multiscale polymer cluster expansion carries an activity parameter κ_k that must contract for the expansion to converge, say

$$\kappa_{k+1} = \frac{3}{4} \kappa_k, \quad \kappa_k = \kappa_0 \left(\frac{3}{4}\right)^k \longrightarrow 0, \quad (14.5)$$

while a physical mass is extracted at each scale by

$$\frac{\kappa_k - \log C_4}{a_k} = c_0 \Lambda > 0 \quad \text{for all } k \geq 0, \quad a_k = a_0 L^{-k}, \quad L = 3, \quad (14.6)$$

with $C_4 > 1$ the polymer coordination constant.

Proposition 14.3. *The two displays above are incompatible. Since $\kappa_k \rightarrow 0$ and $\log C_4 > 0$, there is a finite*

$$k_* = \left\lceil \frac{\log(\kappa_0 / \log C_4)}{\log(4/3)} \right\rceil$$

with $\kappa_k - \log C_4 < 0$ for every $k > k_*$; beyond that scale the cluster sum $\sum_n e^{-(\kappa_k - \log C_4)n}$ diverges and the expansion does not exist. Setting $C_4 = 1$, which discards polymer entropy, does not repair it: then $\kappa_k / a_k = (\kappa_0 / a_0)(3L/4)^k = (\kappa_0 / a_0)(9/4)^k \rightarrow \infty$, so (14.6) cannot equal a finite constant either.

The content of Proposition 14.3 is that a contracting activity and a scale-independent physical mass cannot both be read off the same schedule. Any repair must decouple them — the mass must come from a quantity that is not the activity divided by the spacing.

14.3 Break 3: the moving-time gap can be exactly zero

The moving-time theorem of §13 extracts a limiting gap from approximate sources and one growing observation time per cutoff. Under the error bound $K_n(t) \leq A a_n^{-p} e^{-mt} + B a_n^r$ and probe tilt $|\alpha_n| \geq c_\alpha a_n^s$, it gives

$$M_* = \frac{m(r-2s)}{p+r} \quad \text{at} \quad t_n = \frac{p+r}{m} \log \frac{1}{a_n}. \quad (14.7)$$

Two ways this returns nothing.

Zero margin. If $r \leq 2s$ then $M_* \leq 0$: the theorem is sharp and it yields exactly zero. Here r is the convergence rate of the coarse-graining background and s the rate at which probes must shrink to stay in the quadratic basin of the Wilson action. No calculation in this program, or to this author's knowledge anywhere, certifies $r > 2s$ for four-dimensional Yang–Mills. The inequality $r > 2s$ is therefore a hypothesis with no current route, not a technical condition.

Finite horizon. If the observation time is capped by the box, $t_n \leq c_{\max} \log(1/a_n)$, the optimum becomes

$$M_{\text{opt}} = \min \left(m - \frac{p+2s}{c_{\text{opt}}}, \frac{r-2s}{c_{\text{opt}}} \right), \quad c_{\text{opt}} = \min \left(c_{\max}, \frac{p+r}{m} \right), \quad (14.8)$$

and $c_{\max} \leq (p+2s)/m$ forces $M_{\text{opt}} \leq 0$. Both collapses are registered checks rather than remarks.

14.4 Break 4: a local probe can be blind to the lightest state

Suppose one measures a plaquette correlator and observes

$$\langle \mathcal{O}_{\text{plaq}}, e^{-tH} \mathcal{O}_{\text{plaq}} \rangle \leq C e^{-M_{\text{obs}} t}, \quad M_{\text{obs}} = \log 4 \approx 1.386. \quad (14.9)$$

This does not bound the spectrum.

Proposition 14.4 (Dark sector). *On $\mathbb{C}^3$ let*

$$T = \text{diag}\left(1, \frac{15}{16}, \frac{1}{4}\right), \quad \Omega = e_1, \quad v_{\text{obs}} = e_3.$$

Then $\langle v_{\text{obs}}, T^t v_{\text{obs}} \rangle = 4^{-t} = e^{-t \log 4}$, while the true first excited energy is $-\log(15/16) \approx 0.0645$, smaller by a factor above 21. The probe has zero overlap with the state realizing the gap.

Consequently a decay rate for local plaquettes is a statement about the gap only after the plaquette family is proved *total* in $\Omega^\perp$. This is why totality is a conclusion, not a hypothesis, in the fixed-spacing band theorem (§12); and it is why a continuum argument may not inherit totality from the fixed-spacing statement without proving it again in the limit.

A second, subtler version of the same failure concerns positivity rather than overlap.

Proposition 14.5 (Osterwalder–Schrader null states). *Let $(x, y) \in \{-1, 1\}^2$ be uniform and let time reflection act by $\Theta(x, y) = (y, x)$. The centered observable $F(x, y) = y$ has Euclidean variance $\mathbb{E}[F^2] = 1 > 0$ but physical norm $\|F\|_{\text{OS}}^2 = \langle \Theta F, F \rangle = \mathbb{E}[xy] = 0$.*

So positive Euclidean variance does not certify that a probe exists in the physical Hilbert space at all. An algorithm that selects probes by Euclidean variance can select states that vanish after reconstruction.

A third version concerns the generator rather than the states: with $H_n = \text{diag}(0, n)$ one has $e^{-tH_n} \rightarrow \text{diag}(1, 0)$ for every $t > 0$, so the limiting semigroup is not strongly continuous at $t = 0$ and no self-adjoint generator exists on the second coordinate. Convergence of Gram matrices alone therefore does not produce a limiting Hamiltonian.

14.5 Break 5: the proved regime and the continuum regime are disjoint

This is the structural obstruction, and the one that most changes what a correct argument must look like.

The fixed-spacing band theorem (§12) holds on

$$|u| \leq \frac{u_*}{10\,022\,400\,000\,N}, \quad (14.10)$$

and the spatial-closure estimates hold for $0 \leq \lambda \leq \lambda_c$ with $\lambda_c \in \left(\frac{1}{73}, \frac{1}{72}\right)$. Both are neighbourhoods of zero in the strong-coupling variable.

The continuum limit lives elsewhere. With $g^{-2}(a) = \beta_0 \log(1/a\Lambda)$ and $u = 2/(g^2(a)N)$,

$$u(a) \rightarrow \infty, \quad \lambda(a) \sim g^{-4}(a) \rightarrow \infty \quad (a \rightarrow 0). \quad (14.11)$$

So the interval on which the theorem is proved and the trajectory along which the continuum is taken have empty intersection:

$$[0, \lambda_c] \cap [\lambda(a), \infty) = \emptyset \quad \text{for all sufficiently small } a. \quad (14.12)$$

Scope 14.6. The fixed-spacing results establish a strictly positive gap at fixed lattice spacing in a neighbourhood of strong coupling. They supply no control on the weak-coupling trajectory. An inference from (14.10) to a continuum mass gap is a regime error, and this program has made and retracted a version of it.

What Break 5 demands of a future argument is a *controlled trajectory*: an estimate that survives continuation in the coupling, not a better constant at $u \approx 0$. Improving the constant 10 022 400 000 in (14.10) by any finite factor does not move the boundary of the difficulty. This is why Problem 6 in §16, and not a sharper fixed-spacing estimate, is the load-bearing open problem.

14.6 Break 6: diffusion time is not Euclidean time

A recurring route derives a spectral gap from curvature: positive Bakry–Émery–Ricci curvature on the configuration manifold gives a logarithmic Sobolev inequality, which gives a spectral gap $\lambda_{\text{diff}} > 0$ for the Langevin generator. The step that fails is the identification of λ_{diff} with a physical mass.

Langevin time is a fictitious parabolic parameter; Euclidean time is a spacetime coordinate of the reconstructed theory. In free models the two are related by $\Delta_{\text{phys}} = \sqrt{\lambda_{\text{diff}}}$, but that relation is a consequence of Gaussian structure and not a general isometry. For non-abelian gauge theory no such correspondence is available here.

The companion obstruction concerns the coarse-graining itself. A scale map built from a gauge-covariant Markov kernel that preserves both reflection positivity and locality is not available: any such kernel either forces the holonomies to commute — reducing the theory to an abelian one — or destroys reflection positivity, so that the reconstructed inner product is indefinite. This is why the scale route actually used in §13 is a *deterministic* pushforward rather than a Markov kernel, and why the two-spin counterexample $\mathbb{E}[s_- s_+] = -1$ is recorded as a proposition rather than as an anecdote.

Scope 14.7. Break 6 does not refute the curvature computations themselves. The Haar Ricci constant and the Wilson Hessian bound are correct statements about the configuration manifold. What is refuted is their use as a route to Δ_{phys} , and the curvature results are consequently recorded in this program as inputs to coercivity estimates rather than as a mass-gap mechanism.

14.7 What survives

1. The exact strong-coupling spectral theory of Part I, in its entirety. None of the six obstructions touches a fixed-order statement.
2. The fixed-spacing infinite-volume Wilson band and its literal source frame (§12), on the interval (14.10), with the scope stated there.
3. The moving-time spectral exclusion as an abstract interface: the implication from its hypotheses to a limiting gap is proved and sharp. Break 3 is a statement about whether its hypotheses hold for Yang–Mills, not about the implication.

What does not survive is any inference from the above to a continuum Yang–Mills mass gap. The program states that plainly, and §16 is the list of what would have to change.

15 The refutation register

A refuted claim is evidence that it was tried. This section reproduces the register of every claim this program made, or inherited from its own sources, and then lost — together with the exact thing that killed it.

The register carries 22 entries in three states. 4 are *falsified*: a counterexample or an exact computation contradicted them. 3 are *superseded*: replaced by a sharper statement. 15 are *resolved*: the apparent conflict dissolved, in most cases because the two sides turned out to be differently anchored quantities rather than competing values for the same quantity.

Three features of how this register is maintained matter to a reader deciding how much to trust the rest of the document.

First, a dispute closes by derivation or it does not close. Both values stay recorded side by side, and no code path is permitted to pick one, average them, or prefer whichever is an exact rational because exactness reads as authority. The fourth-order off-axis coefficient closed that way: every cluster of the rotation record reproduces the historical pipeline's own number, and the single cluster that does not is explained ordering by ordering (§6). Its two recorded values remain in the registry with their verdicts attached.

Second, a resolved entry is not a win. The largest class here is resolutions in which a contradiction turned out to be a naming collision — two differently anchored coordinates both called m_4 , for instance. Those cost real time and produced no mathematics, and the register's response was to forbid the ambiguous name rather than to record a victory.

Third, this program has retracted a mechanism of its own. The *tier collapse* — the observation that two of the four fourth-order shape coefficients vanish at every solved rank — was proposed here as a kinematic degree bound with a falsifiable prediction, and the mechanism was withdrawn a fortnight later when one projection turned out to be uncounted (§6). The withdrawal is a numbered decision record rather than a silent edit, because the next attempt at that mechanism needs to know which step failed. The reformulation survives; the mechanism and its prediction do not.

That is a different object from the degree bound of §8, which was never retracted — it was made unnecessary. The two are easy to confuse and this document keeps them apart deliberately.

Falsified (4)

Falsified — killed by a counterexample or an exact computation. These are the four claims this program asserted, or inherited, and then lost.

- **C6 F-2 mobility rule.** The promotion $r_{\text{physical}} = w_{\text{min}} - 2$ is falsified by the pentagonal isotropic cap hop at $r=4$ with $w_{\text{min}}-2=5$. The scoped bound $r \geq w_{\text{min}} - 2$ survives, with survival gates.
- **C7 Stranded-flux zero backend.** Claimed Haar zero for all 20 pentagonal histories refuted by exact Weingarten/network contractions (8/8).
- **C18 Center-only circuits are dynamically dark.** Refuted by the exact fifth-order determinant correction $\delta c_{5,\text{det}} \neq 0$
- **C19 Global fixed-window firewall.** Disproved by the Bernoulli no-go; replaced by the rooted conditional calculus

Superseded (3)

Superseded — replaced by a sharper statement. The original is not wrong so much as no longer the best available.

- **C5 String-tension sign at $O(u^3)$.** $-61/408$ (the earlier $+61/408$ was an odd-order convention error)
- **C11 Compact-gap constants.** Corrections: $c_0 = -3(N^2-3)/(16N)$; radial moment difference $3N^2+1$; dilation $\text{eps}^4 = N/(2*\beta)$;
- **C12 Curvature misreads.** Two-stage. The kappa values are RADIAL DIRECTIONAL curvatures, not Hessian entries at Gamma; a genuine Hessian needs $\beta = 2*\alpha$, violated by the

historical kernel.

Resolved (15)

Resolved — the collision was dissolved. In most of these the two sides turned out to be differently anchored quantities rather than competing values.

- **C1 Fourth-order rest scalar — dissolved, an anchoring distinction.** NOT a contradiction. $q_{\text{band}}^{(4)}$ and $m_{\text{Gamma}}^{(4)}$ are not two competing estimates of the same scalar coordinate: the first is a band-kernel anchor, the second a vacuum-subtracted physical Gamma-point coefficient. They are differently anchored coordinates related by a translation-local scalar shift, and such a shift cannot change the centered operator, its eigenvectors, the SOS factorization, the mobility coefficients, or the bandwidth.
- **C2 Fourth-order off-axis coefficient C_{shp} — resolved by derivation 2026-09-04.** $C_{\text{shp}} = C_{\text{SHP_HISTORICAL}} + 25/1024 = -13035490122347/550663802582400 = -0.0236723$, the cluster-assembled coefficient read in the kernel's own basis (ADR 0024). Both recorded kernels are wrong on the rotation amplitude: the historical by the sixteen orderings of the adjacent-face cube completion its own Stage-3I ledger lacks (8 of 24, each with the right weight; the shortfall is exactly $25/512$ on ρ), the cold dump on u , π and ρ alike.
- **C3 Status of the fourth-order "theorem".** Newest status controls. The §5.2 values are sealed; everything else is $C1/C2$.
- **C4 Coupling conventions.** The archived $Y = 2*\beta/3 = 4u$ line is a definition/label error, not a rescaling rule. Printed coefficients were already in $u = \beta_{\text{lat}}/6$. Never apply $4**n$ factors to them.
- **C8 Pentagonal degenerate space.** $E_{\text{cap}} = 10/3 \neq E_{\text{side}} = 8/3$; choose the physical degenerate eigenspace first
- **C9 SU(4) exceptional correction.** Only the flat-branch eigenvalue identity holds, not scalar-ity of the matrix
- **C10 Determinant sectors shift only q_{N} .** True for $N \geq 4$, false at SU(3): $\Delta_{\beta_3} = -25/64$ and $\Delta_{q_3} = -16863189551/76406976000$.
- **C13 C-even second-order hopping.** $-481/612$ superseded by $-11/306$ (the vacuum-mediated route was omitted)
- **C14 Betti "+2".** Reinterpreted as $b_2 - b_3$; the open box has no extra states
- **C15 Tetrahedral local coefficient.** Re-derived 2026-08-21 (tetrahedral Haar-resolvent suite): the asserted $-8/(N(N^2-1))$ is exact at symbolic N . The derivation reconstructs the primitive law's three stated ingredients — $1/N$ per merge from the fundamental-pair Haar moment (independent of shared-path length, derived not assumed), the certified electric convention $E(L) = L*C_F/2$, and exhaustive temporal orderings — and reproduces every certified or printed instance first (sealed-core cube $-5/48$ with the $(-8,-8,-4)$ temporal classes, prism 64 with its cap-sector 24 dissolved as a sector ...
- **C16 SU(5)/SU(6) fourth-order completeness.** The payloads were shipped after all, and are now machine-checked (restored-payloads suite, 2026-08-21): the SU(6) determinant certificate re-derives $\Delta_{q_6} = 6/343$ exactly, momentum-independent with A , B , and the bandwidth

unchanged, and the SU(5) stage-1 summary pins the 895,524-pair scan with zero determinant sectors. Two upstream word manifests remain absent (their SHAs are recorded in the certificates), so the evidence level is output-certified, not cold-reproduced.

- **C17 Fixed-rank holdout overclaim.** Corrected: q_N matches stored values for $N=7..18$, but full kernels only at $N=7,8$; the A/B sample ledger remains open.
- **C20 RUN15 internal display artifact.** Cosmetic; the gate value is the intended exact one
- **C21 Monte Carlo gate integrity.** `next14.json`'s "23/23 gates" is downgraded: one gate is a literal truth value, the JSON is not source-hash bound (non-RFC NaN tokens), and no raw ensemble exists. Numbers stand as structured numerical evidence; the certificate claim does not.
- **C22 Gate-85 equality.** Not a numerical contradiction but a status correction: the final diagonal shift was chosen to produce exactly that equality, so gate 85 certifies internal bookkeeping, not independent agreement.

15.1 Claims withdrawn by this program

Distinct from the register above, the following are claims this repository itself advanced and then withdrew. They are listed separately because their failure mode is different: nobody contradicted them, the argument was found not to close.

- **The tier collapse as a kinematic degree bound.** Proposed with a falsifiable prediction — that the next shape tier would first appear at sixth order — and retracted eleven days later. The count bounded the vertices of H_4 ; the carrier projection $d^\dagger(\cdot)d$ contributes one further power of the Bloch variable that the argument did not carry. Recorded as `docs/decisions/0005-retracting-the-degree-bound.md`. The reformulation stands, the mechanism does not, and the collapse is now established by the range-one argument of §6 instead.
- **An unrestricted exponential-source radius.** Inferred from independent bounded sources; the inference does not hold, and the reformulation is Problem 7.
- **Closure of the source-energy transport budget.** Claimed complete, rejected on review: a vacuum-only skew generator does not control a source that the generator transports rather than annihilates. The corrected partial results are retained and the obligation is Problem 2.

The falsified entries above — the mobility rule, the stranded-flux backend, the dark centre-only circuits and the global fixed-window firewall — are a different category and are not repeated here: each was killed by a counterexample or an exact computation rather than by an argument failing to close.

Each of these was, at some point, written down in this program's own documents as established. That is the reason the machine-verification tier of §2 is computed rather than assigned, and the reason the present document quotes its counts from the ledgers by macro rather than by hand.

16 The obligation register

This is the operative section of the document. It states, as numbered problems, what would have to be proved; and it then reproduces the repository's full route register, including every route that is dead and the reason it died.

The numbered problems come first because they are curated: they are the routes this program judges to be live and decisive. The register that follows is not curated — it is the whole record, 111 routes across 20 open and 5 discharged gaps, of which 64 are complete, 20 live, 12 untried and 15 dead.

16.1 The problems

Ordering expresses research judgement. It is *not* a claim that an earlier problem is a prerequisite for a later one; the ledger records explicit inputs separately, and each problem below states its own.

Problem 1 (Conditional-score domination, uniformly in the coupling). For the synchronized transport of the four-face $SU(2)$ conditional model, establish

$$K_g(w) \leq C_0 + C_1 W_g(w) \quad \nu_g\text{-almost everywhere,} \quad (16.1)$$

with C_0, C_1 independent of g as $g \rightarrow 0$, retaining the chart-complement, rare-fiber and antipodal contributions.

Available. The actual source moment; the interval-obstruction repair; conditional-score continuity; synchronized conditional dilation; the antipodal Hessian spectrum, gauge-normal coercivity, angular reduction and gauge-score invariance — all proved. Exponentially small complement suppression outside a neighbourhood of the complete minimizing set, and a uniform Agmon action gap on the complement, are proved.

Missing. Polynomial relative-amplitude control; the tube moments; the actual normal/angular comparison.

Consequence. With the established source-potential moment, (16.1) makes the uniform weighted score bound applicable. It does *not* supply source-energy transport or interacting-grid uniformity; those are Problems 2 and 3.

Next computation. For the synchronized field, obtain the actual conditional tube moments and the differentiated amplitude bound; then use the established antipodal magnetic normal and angular geometry to prove the true-ground score estimate, controlling the outside deviation against the same reference. Compare the resulting complete variance with $1 + g^{-2}\mathbb{E}(V \mid w)$, or prove a sufficient source-weighted pairing directly.

Dead here. Inferring uniform domination from endpoint requirements on arbitrary cutoffs; differentiating a global fixed-rank physical Coulomb source projection across flat holonomies; bounding the global true-ground Fisher matrix by $C_0 + C_1 \sqrt{u} v(U)$.

Problem 2 (Source–vacuum transport energy jets). Bound the complete compact source and vacuum transport on a common energy domain against a physical clock. A vacuum-only skew generator is insufficient: the source generator transports the ground state, it does not annihilate it.

Available. The ground jets; the ground-frame generator; the score–Poisson identity; the conditional derivative; H_4 derived from H_0 ; the projection-jet recurrence — all proved.

Next computation. Decompose the complete generator into vacuum and conditional-source parts; retain commutator and energy-domain terms through two derivatives; test the proposed powers of g on the full energy graph rather than only on ground vectors.

Recorded failure. An earlier closure of this problem was claimed and rejected on exactly the ground that the vacuum-only generator does not see the source. The rejection is preserved; see §15.

Problem 3 (Interacting-grid comparison with uniform constants). On the actual coupled Wilson grid, with the true vacuum, the full fast space, source metrics, interfaces and a common physical clock, prove a comparison whose constants are uniform in the number of interacting blocks.

Available. The literal coarse form; covariant block sources; the form Schur scale comparison — all proved.

Next computation. Two genuinely coupled adjacent blocks. Compute the full conditional and source form including the interface and the high retained directions, and isolate a local estimate whose constants sum without a block-count factor.

Why uniformity is the whole problem. Fixed-block estimates are available in quantity. A constant that degrades with block count telescopes to nothing, which is the same arithmetic failure as Break 14.1 in a different variable.

Problem 4 (The signed conditional defect in the actual vacuum). On compatible finite periodic SU(2) blocks with the actual positive Wilson vacuum, bound the full signed conditional reference defect, retaining all localization and domain terms.

Available. The first and third residual estimates, proved.

Next computation. On the smallest coupled block, derive outside-pressure and projector derivatives alongside the magnetic and metric terms; check the Gaussian cancellation exactly; then bound the remaining full expression with its cutoff commutators.

Problem 5 (Summable normalized RG and source increments). For the actual normalized RG map and the renormalized observables after marginal matching, with all measures and source spaces identified, prove a cross-scale increment bound with scale-uniform remainder constants that is *summable*.

Available. Contractive drift on a common space, proved.

This is Break 14.1. The required improvement is from $O(g_k^2)$ to $O(g_k^4)$, and (14.4) shows the normalization supplies no small factor, so the improvement must come from an exact cancellation of the leading partition-function covariance.

Next computation. Work out one genuine normalized blocking step with its observable transport; isolate the marginal contribution; and calculate the surviving common-norm increment before claiming a power-improved bound.

Problem 6 (A controlled trajectory out of the proved interval). Connect the fixed-spacing construction of §12, proved on $|u| \leq u_*/(10\,022\,400\,000\,N)$, to the weak-coupling trajectory along which the continuum limit is taken.

This is Break 14.5, and it is load-bearing. The two regimes are disjoint. Borel–Padé consistency of the strong-coupling series is not such a trajectory: agreement of a series with a resummation is not continuation of an estimate.

What a solution looks like. An estimate whose constants survive continuation in the coupling, or a second construction valid at large u together with a proof that the two constructions describe the same theory on an overlapping window. The second route has a named prerequisite already discharged in part — the dimension-five operator irrelevance lemma — and the coupling window between the two constructions is the object to close.

Not a solution. Any improvement of the finite constant 10 022 400 000.

Problem 7 (A valid source-family formulation). Identify the exact source operator, normalization and norm consumed by the selected spectral comparison, and derive a local, connected or normalized replacement *together with* a proof that the replacement preserves that comparison and a nonzero physical source weight.

Refuted approach. Inferring an unrestricted exponential-source radius from independent bounded sources. Linear susceptibility bounds do not bound the centered exponential source.

Available. The raw α -dependent partition bound for bounded sources, established. Scalar Kotecky–Preiss inequalities and raw moment estimates remain useful ingredients but construct neither the tilted activities nor a gap.

Next computation. Compute the candidate source norm and its spectral normalization in the explicit independent product law; compare with $(M(2\alpha)^n - M(\alpha)^{2n})/n$; then state the precise interacting inequality that remains.

Problem 8 (One complete sixth-order cluster). At fixed $SU(3)$, on one smallest admissible connected plaquette support, evaluate the eighteen electric folds with exact Haar and Gram quotients, physical projections, electric denominators and rooted subtraction.

Available. The Hodge–Feshbach splitting; the fourth-order support theorem; the Hermitian folded support; local sixth-order non-cancellation; the one-face sixth order — all proved (§9).

Consequence. The extra rational band-shape component is already forced under the local-shell and third-order hypotheses. Computing the direct term determines its independent polynomial shapes and coefficients, and decides whether the local structure survives. It changes neither the fourth-order support theorem nor the global sixth-order census.

Next computation. Evaluate the eighteen folds on a connected multi-plane support with all Haar weights and rooted subtractions. Reduce the direct term separately; then combine it with the known mixing numerator at common denominator, retaining both the direct polynomial shape coefficients and the invariant remainder.

Unlike the others, this one is finite and priced. It is the cheapest decisive computation on this list.

16.2 The full route register

What follows is generated from `ledger/gaps.yaml`. Dead routes are included deliberately and at length. A route marked *cannot decide* is structurally unable to resolve the named claim, which is a stronger statement than having failed to.

G1 — Ship or restore machine-readable payloads that are only record-backed

Tier 0 (days); partial.

$SU(3)$ 189-record kernel with reference SHA; Q_32 and P_402 ledgers; $SU(5)/SU(6)$ exceptional certificates;

Ledger status. largely discharged 2026-08-21 — three of five families were shipped all along and are now machine-checked (restored-payloads suite): the 189-record kernel with both reference SHAs, the Q_32/P_402 Newton ledgers, and the $SU(5)/SU(6)$ certificates. Still absent: the pentagonal dual-cold bundle (manifest row A60 records the SHA any restoration must hash to) and the tetrahedral certificate, for which no reference SHA exists anywhere — that one was G5’s to close, not this gap’s, and G5 closed it by re-derivation on 2026-08-21 (the artifact itself remains absent). A...

G25 — The charge-even sector is registered and unchecked

Tier 1 (weeks); partial.

What remains. The suite reaches orders 2 and 3, which is where the corpus's C-even ledger stops. Fourth order is untouched on this side: the pre-registered $O(u^4)$ criterion in the corpus is a statement about C-odd flatness, and no C-even fourth-order coefficient is registered here at all, so the sector's fourth order is not T3 – it is absent.

Ledger status. opened and largely discharged 2026-08-28. The C-even sector was registered and inert: LEAK_2, LEAK_2_EVEN and LEAK_3_EVEN were read by no check body, band_
assembly() was dead code, and the only C-even geometry the repository checked was two hand-written matrices at Gamma and R. The 'charge-even band, exactly' suite (15 checks, all T1) closes the second- and third-order half. Both entries of constants.DECLARED_COINCIDENCES are now checked rather than only declared: -11/306 is ell_N...

Depends on. G3

Live routes.

- **fourth-order domino, the U4 falsifier.** PROGRESS 2026-09-12, part (2) DONE: conn_vac is derived and no longer T3. The linked two-face vacuum weight is now computed exactly by two routes that agree (swap-odd suite, new T1 check). At SU(3) the third engine's loop calculus gives the two-face vacuum sector 0, 0, -3/2, 9/16, -54321/837760 against twice the one face, so the linked cumulant is 0, 0, 0, 0, -327/83776: the corpus's omega4, identical for the coplanar and the perpendicular pair, with the linked $O(u^2)$ and $O(u^3)$ weights exactly zero as the v10a.7 gates reported in floats.

G4 — Pentagonal $O(u^4)$ closure in-corpus

Tier 1 (weeks); partial.

Fierz closure of the 20-history cut spaces, Gram quotient, physical resolvents, summed compression.

Ledger status. partial — the raw history Gram/Haar frontier is closed (8/8 gates, cold rerun in settlement/); the Fierz closure, physical quotient, and summed resolvent compression remain (see progress.remaining).

G6 — B_N fixed-rank holdout ledger for N = 7..18

Tier 1 (weeks); open.

G7 — Native torelon reruns of sigma_5 and sigma_6

Tier 1 (weeks); open.

Replace the KPS historical targets.

G8 — Cold one-shot regeneration of the all-rank fourth-order bundle

Tier 1 (weeks); open.

Promotes output-certified to cold-certified.

G10 — Full fifth-order pentagonal coefficient

Tier 2 (months); open.

Support-resolved folds for the 572 return histories before rooted subtraction.

G11 — Interval rigor and near-Gamma touching gates

Tier 2 (months); partial.

Outward-rounded pi. Without this, a fixed-momentum coefficient theorem must NOT be promoted to a uniformly isolated near-Gamma band theorem: for $|k| \lesssim u$ the $O(u^2)$ splitting $\sim u^2|k|^2$ meets the $O(u^4)$ scale.

Ledger status. partially discharged

G12 — Hyperhoneycomb $O(u)$ candidate

Tier 2 (months); open.

Tests the framework beyond cubic cells.

G9 — m_6 : global sixth-order geometry census

Tier 2 (months); open.

Contraction with external-memory sharding; folded weights and carrier census pre-cleared.

Depends on. G3

Live routes.

- **Assemble one complete sixth-order cluster and test its extra carrier-shape component.** The September 11 continuation derives all 18 universal Hermitian electric folds, evaluates the complete one-face $SU(3)$ rooted coefficient $-2055143/35123200$, and reduces all 25 H4 mixing pairs. Under the recorded third-order factorization and local-shell hypotheses, an additional rational shape cannot cancel against any local direct H6: $q^3 \lambda_6 \bmod q^2$ is $4 \kappa a e^2 - 3 \kappa q e^3$ for symmetric polynomial direct F. The remaining task is the direct connected multi-plane H6 evaluation, including its Haar/Gram quotient, electric...

G13 — Classification theorem for shortest physical temporal histories

Tier 3 (unbounded); open.

Determine r_{escape} from the energy-decorated incidence/Feshbach graph without exponential enumeration. Would subsume tetra/prism/cube/pentagonal/ hyperhoneycomb as one theorem, and includes proving partition-independence of the linked/folded assembly.

G14 — Fourth-order support mechanism and its higher-order extension

Tier 3 (unbounded); partial.

Derive the actual supported operator words from complete physical histories, with the Haar quotient, electric resolvents and folded assembly retained. Use the established fourth-order support theorem as the baseline for a bounded sixth-order calculation. The Hodge splitting alone does not assert an all-length word classification or a dynamical sixth-order coefficient.

Ledger status. Fourth-order tier collapse is established by `RESULT:TIER_COLLAPSE_ACTUAL_H4_SUPPORT`: the recorded kernel has support $\{I, U, S, S^2, R\}$, with cleared carrier

span $\{q, q^2, e_2\}$. The broader R-degree inference is falsified by $\sigma(\text{UR}) = -2q e_2$. The open successors concern history-to-word support, higher multiplicities and the separate proper-return identification in other geometries; the September channel computations are established inputs.

Live routes.

- **prove channel-wise universality from the Haar contraction – that a channel’s value depends on which links are doubled and in which irreps, not on where on the cluster they sit – for u, the fan and the single-contact dressing.** Opened 2026-09-04 (ADR 0029). With every channel an identity in $Q(N)$, the mechanism of the tier collapse is a statement about the channels alone: B is unpopulated by the carrier projection, $D = 0$ is $u^2 = 2u$, and $u^2 = 2u$ is the L chain equalling the straight chain channel by channel. What is left is a proof of that equality from the contraction rather than a computation of it.

Untried routes.

- **derive the actual fourth-order Hodge support from the histories, including the absence of UR, RU and repeated R insertions.** Reviewed 2026-09-10. The narrower actual support $\{I, U, S, S^2, R\}$ already proves the recorded kernel’s $B=D=0$. Explain this support from the complete electric/Haar and folded histories, including the absence of UR, RU and repeated R insertions;
- **test the Hodge-summand mechanism in a second geometry by computing the tetrahedral Q-projected proper-return contributions, the first clause of U7’s falsifier.**
- **audit the premise that $D - (1/2)\{H_2, V_2\}$ is the Hermitian fourth-order effective Hamiltonian on a cluster with $PVP = 0$ (the pair element at $N \geq 4$), against a direct Bloch or Schrieffer-Wolff expansion on the two-face cluster at $N = 5$.** Taken as given by every lens of the 2026-09-04 adversarial verification of ADR 0027; unrefuted, unaudited.
- **audit the rank-generic resolvent at $N \geq 5$ – link_spectrum roots, the dropping of the E0 component, vacuum intermediates – against the SU(3) Casimir table route continued to $N = 4, 5$ by hand.** Taken as given by every lens of the 2026-09-04 adversarial verification of ADR 0027; unrefuted, unaudited. Partly covered later the same day (ADR 0029): over $Q(N)$ every eigencomponent of every resolvent is verified to be an exact eigenvector of every single-link H_0 with the SU(N) Casimir energies (21,556 components), so the spectra are no longer read off characteristic-polynomial roots at all;
- **the completeness of the eleven-cumulant decomposition of β_N at general N rests on the charge-counting argument alone (a connected four-insertion history from P to Q is $P \bar{Q} X \bar{X}$, the pair, or the cube); enumerate the fourth-order support at $N = 5$ from the engine and compare with the 189 Hodge-form keys.**
- **the $N = 4$ row of the beta record rests on the determinant trick (pseudoinverse Weingarten at $n > N$ plus the epsilon insertion) in the three-face cumulants; audit that path at $N = 4$ against an independent epsilon-network reduction, and make the (4,0)-family pair element feasible the same way.** Sharpened 2026-09-04 (ADR 0029): the closed forms derived over $Q(N)$ are regular at $N = 4$ and agree with the $N = 4$ row of every per-rank record, so whatever the determinant trick contributes there is what the generic rational function gives. The symbolic run also names the $N = 4$ degeneracy: the state P^3 with every link of the plaquette in $\Lambda^3 F$, which is $\bar{F}$ at $N = 4$, has energy E0 exactly there (the one resolvent denominator with an integer root ≥ 4). The audit of the path itself is still open.

Dead routes, and why they died.

- **derive why the fourth-order dynamics is R-linear – why no four-insertion history inserts the cross-plane operator twice – which is what the tier collapse now reduces to.** The proposed reduction is false: UR contains one R but its symbol $-2q e_2$ populates the B tier. The actual fourth-order collapse instead follows from support $\{I, U, S, S^2, R\}$. The step name is retained to preserve its existing route id;

Completed routes (2). read the mechanism off $u(N)$ – decompose the two-hop cumulant by intermediate irrep and match its denominator factors to resolvents; the same channel decomposition for the corner and fan dressings of the rotation orbit.

G15 — Symbolic-N Gram transcript

Tier 3 (unbounded); open.

Promotes c_1, c_2 to unrestricted fixed-N theorems; one more order of the equivariant harmonic-well lemma.

G16 — Rank-uniform control in $\tau = \beta/N^3$

Tier 3 (unbounded); open.

Neither fixed-regime series supplies an overlap theorem. PROGRESS 2026-09-11 (ADR 0046, runs/planar_band_2026-09-11, suite "the planar limit of the fourth-order band (G16)", five T1 checks): the strong-coupling side is now exact through fourth order. Every fourth-order cumulant of the β_N assembly is $O(N^{-7})$ in both sectors while its resolvent channels are $O(N^{-3})$, the N^{-3} and N^{-5} channel totals vanishing identically in all sixteen cluster/sector pairs;

G17 — Source-family stability and a valid source-radius formulation

Tier 3 (unbounded); open; load-bearing.

Identify the exact source operator, normalization and norm consumed by the selected spectral comparison. Derive a local, connected or normalized replacement only with a proof that it preserves that comparison and nonzero physical source weight. Linear susceptibility bounds do not bound the centered exponential source.

Ledger status. The raw α -dependent partition bound is established by DERIV:TRACK_A_SUPPORTED:G17_P with $K_\alpha = \exp(|\alpha|/2)$. The unrestricted centered exponential-source radius is not a consequence of clustering: DERIV:TRACK_A_SUPPORTED:G17_I gives divergence even for independent bounded centered sources. A valid source-family formulation and its actual interacting estimate remain open. The fixed-spacing infinite-volume Wilson band and literal-source...

Untried routes.

- **Reformulate the source family while preserving its downstream spectral comparison.** Choose a precisely stated local, connected or normalized source formulation and prove the required relationship to the actual spectral comparison, including nonzero physical weight. Test it in the independent product model before seeking an interacting estimate. This is a candidate route with a source-identification obligation, not a claim that the original unrestricted exponential radius can be recovered from covariance summability.

Dead routes, and why they died.

- **Infer an unrestricted exponential-source radius from independent bounded sources.** DERIV:TRACK_A_SUPPORTED:G17_I constructs an independent bounded centered source whose exponential variance divided by footprint cardinality tends to infinity. This refutes the stated inference, not every localized or normalized source formulation.

Completed routes (1). Establish the raw alpha-dependent partition bound for bounded sources.

G19 — Continuum limit

Tier 3 (unbounded); open; load-bearing.

The actual infinite-volume Wilson construction and complete physical source band are now established at fixed spatial spacing on a common small- u interval. A controlled scale trajectory connecting it to large u , renormalized continuum correlations and a positive physical mass gap remain open. Borel-Pade consistency is not that trajectory.

Ledger status. Reconciled 2026-09-09 after reviewing 43 distinct G19 document versions. Non-linear finite-cell shells and scores, complete Gaussian fast/source bounds, fixed-spacing matching, and conditional continuum/OS transport are retained. The newly cited Balaban continuum proof stops at (7.6): its $O(1/k)$ expectation increments do not imply the Cauchy conclusion in Theorem 7.1, which invokes a nonexistent $O(1/k^2)$ estimate. Its (6.10) weight $\kappa_k = (3/4)^k \kappa_0$ also fails the positive entropy margin required in (8.14).

Live routes.

- **Prove the actual conditional-score domination M10.** The source-potential moment M8-M9 is established. The open target is the actual small- g conditional score $K_g \leq C_0 + C_1 g^{-2} E(V|w)$, including rare fibers, with uniform nonnegative C_0, C_1 . This sufficient route implies the weighted score/Hardy estimate M11-M15;
- **Control complete source-vacuum transport energy jets R10.** Raw H_g derivatives and vacuum-only ground jets are established. Prove the complete source/vacuum generator bounds R10, $\|A^\wedge(r)(g)\|_{(q_g \text{ to } q_g)} \leq d_r g^{-(r-1)}$ for $r=0,1,2$, or use the source's direct mixed-form R11 alternative. These are alternative sufficient entrances to the complete residual budget, not cumulative prerequisites.
- **Control the interacting quantum comparison in redundant compact coarse variables.** Use the actual smooth matrix observation with its exact coarse gauge action and boundary source lift. Prove interacting true-vacuum marginal and full complementary energy estimates, including nonflat localization, noncommuting compact slow dynamics, the complete direct/exchange/metric/source correction and high retained tails. The ambient redundant projector is not a physical quantum Schur projection.
- **Bound the nonlinear excess above the exact quadratic memory on complete physical source windows.** Combine the established dynamic cubic energy and exact full fast Green identity with quartic magnetic, electric metric, Haar, moving-source, baseline cross and fast-form variation terms. Prove bounds for the actual spatial coefficients and the complete interacting remainder with retained harmonic localization, actual compact dynamics or explicit cancellations. Full source algebra, high retained directions and a summable physical-clock trajectory remain open.

- **Transport the endpoint and localized true-ground estimates through actual interacting scale blocks.** The additive complete endpoint and selected static-score comparisons are established inputs with different scopes. Prove uniform actual interacting fast/source or memory estimates and compare the resulting endpoint law with a consistent coarse quantum theory in one physical clock. For a static Schur route, retain its high-source coupling rather than inferring a full gap from the chosen chart.
- **Connect the actual Wilson fixed-scale construction to a controlled continuum trajectory.** Actual fixed-spacing infrared transfer and sources remain the controlled endpoint. The ultraviolet side now has a general fixed-cell physical first-cluster theorem and all-size planar and three-dimensional harmonic interface bounds, with planar low-mode/source frames and retained periodic harmonic directions. Gaussian Schur elimination retains memory and induced mass, giving a quantitative sorted-frequency approximation without small interface coupling.
- **Control the true interacting Wilson score and forms on the relevant energy space.** The exact $SU(2)$ central true-ground Fisher matrix grows at least linearly in u , so the proposed global $C0+C1 \sqrt{u}v(U)$ estimate is no longer a live target. Prove an actual full- Q energy weight and a resolvent-weighted or coarse-gradient-energy score estimate on the relevant source window. A good-region small ratio plus controlled bad-region gradient-energy leakage yields a conditional small window Schur loss.
- **prove actual interacting W6 and coarse/source matching through spatial refinement.** The exact selected inverse-pairing and compatible source transport theorems remain available. R4a supplies a flat conditioned Gaussian reference budget; R12a identifies its zero complete pressure defect. *Cannot decide: G19.*
- **supply the dimension-five irrelevance lemma, then close the coupling window between the two constructions.** The two rigorous constructions in this repository approach the continuum from opposite ends of the coupling axis, and the target is now the interval between them. From the strong-coupling side SC17 and BA20-BA25 give a volume-uniform physical gap for the actual vacuum, and the thermodynamic continuation gives an infinite-volume vacuum with a closed generator and spectrum in $\{0\} \cup [4\epsilon/3, \infty)$, for k/ϵ at most $\lambda_{c_}$, the root of $6561 l^4 + 1458 l^3 - 81 l^2 - 72 l + 1$ in $(1/73, 1/72)$. From the...
- **Extend the explicit square current to the complete finite-coupling residual.** Prove the complete transported finite- g current remainder against the actual interacting energy, with reference source synthesis and the required uniform parameter bounds. The no-diagonal criterion then supplies the selected-inverse estimate. Compatible source domains and metric transport remain explicit, and coupled-volume extension is a further step.
- **Transfer a physical continuum gap from one growing-time estimate per cutoff and approximate sources.** The original spectral-measure theorem and sharp rate budget are proved. The compatible-kernel successor now constructs the limiting physical semigroup and total centered defining space from finite Gram limits and fixed-time compatibility; positive-time divisibility supplies zero-time continuity, so a separate vague-limit or frame estimate is then unnecessary.

Untried routes.

- **Establish the actual interacting-grid fast and source comparison with uniform constants.** Construct the true interacting-vacuum fast floor and complete local centered/source comparison uniformly in block count, retaining interfaces, Gauss constraints, nonreducing source

directions and the normalized coarse operator. A fixed-square bound, additive tensorization or harmonic floor alone does not prove this target. The chosen physical scale trajectory and summable errors must then satisfy the closed-form iteration theorem’s actual hypotheses.

- **Bound the complete signed SC17 conditional defect in the actual vacuum.** Instantiate the actual uniform smallness input for the Hessian defect $D_B = \text{Hess}(V_B + (H_out - \Omega)/\Omega) - 2\epsilon A^2$, with pressure, metric, projector, moving-source and cutoff/product-domain terms assigned once. Preserve signed cancellations before bounding the complete expression. The conditional Riccati and block-geometry consequences are available, and the thermodynamic/physical-time constructions already stand at fixed spacing;
- **Prove normalized RG and source increments with a summable common-space remainder.** `RESULT:COMMON_SPACE_CONTRACTIVE_DRIFT` supplies the proved abstract convergence criterion; its actual Wilson map and source hypotheses remain open. Construct the actual RG transformation and matched normalized observable maps on common spaces, subtract the running marginal terms, and bound the remaining expectation/source increments by a summable sequence.

Dead routes, and why they died.

- **Infer actual uniform M10 from the endpoint requirements on arbitrary Q8 cutoffs.** A permitted smooth radial cutoff freezes a fixed small regular Q while moving its actual conditional center. Full conditional concentration and a finite-flow speed contradiction rule out M10 for that choice. This is an existence counterexample within Q8, not a proof against every admissible transport or the synchronized successor.
- **Differentiate a global fixed-rank physical Coulomb source projection across flat holonomies.** An explicit SU2 stabilizer change gives a norm-one projection jump. The directions are gauge off identity and physical harmonic at identity, so this does not refute the uniform fast bound. Keep the smooth redundant actual observation and impose Gauss exactly instead of differentiating the changing Coulomb quotient. *Cannot decide:* G19, G23.
- **Bound the global true-ground Fisher matrix by $C0 + C1 \sqrt{u} v(U)$.** The exact actual SU(2) bouquet gives weighted Fisher at least $(64/3)u - O(\sqrt{u})$ at $U = -I$, where $v(U) = 4$. Smoothness makes the violation persist on open neighborhoods, so uniform almost-everywhere variants fail. The generic intrinsic-score and full-gap theorem remains valid under its explicit hypotheses, and other weighted or energy-restricted score estimates remain available. *Cannot decide:* G19, G23.
- **Infer a scale contraction from the forward averaging derivative and a uniform raw fiber diffusion gap.** The preserved EX-014 Section 1 STEP 6 reverses the pullback derivative: a linear arithmetic-average test requires reverse gradient constant m , not $1/m$. Correct quotient metrics remove the artificial gain. The actual Wilson horizontal score also has an order- g geometric term in a physical two-holonomy example; *Cannot decide:* G19, G23.
- **Griffiths-type monotonicity of the electric-flux sector free energies in beta.** proposed 2026-09-03 (Alex): a correlation inequality that holds at every beta and makes the electric-flux free energies $F_e(\beta)$ monotone in beta, sector by sector – the sectors are Z_N -valued, so the non-abelian obstruction to Griffiths’ inequalities for Wilson loops need not apply – would carry the gap from strong to weak coupling the way Griffiths carries Ising order across phases. Closed the same day by a sign, before the inequality itself was attempted: the Ising argument needs the

monotone quantity to be INCREASING in beta, and the certified series says... *Cannot decide: G19.*

- **compare the whole compact Wilson fast form from above by a Gaussian form after finite-rank retention.** W1-W3 use actual compact SU2 spectral growth and finite-codimension min-max to refute this implementation at every fixed positive coupling. Compact eigenvalues grow quadratically in the irrep label while the Gaussian radial reference grows linearly. This does not refute a compatible compact reference or selected-inverse comparison. *Cannot decide: G19.*
- **infer a regulator-uniform scalar single-site angle margin from the full square-root Gaussian precision.** For $M_m = \sqrt{L+m^2}$ on a periodic scalar lattice, $(L+m^2)^{-1} = m^{-2}$ gives $M_m^{-1} = m^{-1}$ exactly, so the row sums are m ; every off-diagonal entry is negative by the heat representation, hence $\sum_{f \neq e} |M_m(e,f)|$ equals $M_m(e,e) - m$ and the conditional maximal-correlation row is exactly $\kappa_m = 1 - m/M_m(e,e)$. Since $M_m(e,e)$ tends to a finite positive constant, κ_m tends to one as m vanishes: numerically .6154 at $m=1$,... *Cannot decide: G19.*

Completed routes (28). Derive the actual conditional continuity score and locate the constrained Agmon centers; Synchronize compact radial dilation with the actual conditional minimum curve; Resolve the antipodal conditional magnetic normal geometry; Control time-integrated conditional cubic energy with retained-mean localization; Identify the actual nonreducing Gaussian fast inverse with its energy prior; Resolve the unlocalized local harmonic cubic exchange target; Keep the exact quadratic endpoint and its complete low-frequency source normalization; Control the entire conditioned quantum Gaussian fiber at each fixed block scale; Determine the first actual Wilson ground and chosen-chart source correction; Extend the actual path-source fast bound over every flat holonomy; Realize the uniform transverse harmonic source as an actual local covariant path observation; Compare the complete actual endpoint spectrum with all retained source directions included; Establish actual local true-ground score loss on complete additive selected source charts; Identify the true-vacuum literal coarse form and additive physical fast space; Lift the full harmonic boundary inequality to the entire Gaussian quantum fast form; Establish the actual two-face vertical, coarse and full-vacuum complementary forms; Retain planar and periodic three-dimensional harmonic interfaces while isolating box-scale fast modes; Derive an energy comparison that retains arbitrary cross coupling and the induced norm; construct actual discrete-time Wilson blocks and cancel disconnected vacuum factors at a fixed observable; control the complete marked shell and onto literal source frame at fixed spatial spacing; sum temporal Wilson matching and transport the carrier with its specified observable overlap; assemble the full spatial Schur excess, second Wilson and source jets and accumulated physical-clock budget; replace the failed global fast-form comparison by an admissible selected-resolvent Schur identity; reconcile the broader G19 corpus and derive the flat reference and Cauchy repair inputs; audit the GPU and Lean closure claims against their actual operators and hypotheses; Construct the complete first residual current for the fixed twelve-edge square; Derive the compatible-kernel reconstruction and moving-time full-space implication; Optimize finite decay channels and time-amplified errors in the physical clock.

G20 — SAFE-region curvature certificate

Tier 3 (unbounded); open.

The notes program's quantitative core: a certified lower bound on the projected physical Hessian H_{phys} over a SAFE ball around the identity (claimed constants $\kappa^* = 0.25$, $\delta = 0.006$,

headline min eigenvalue 0.248), feeding curvature $\rightarrow$ LSI $\rightarrow$ gap. As reviewed, the headline is UN-REPRODUCED: the sole source table is contradicted by the archive’s own scans and independent recomputation (single-link Haar profile is flat at $1/4$), the operator’s code has never been executed, and no lattice size is stated anywhere. What discharging requires: run the imported...

G21 — Davies/Combes-Thomas $O(m)$ decay for the massive Maxwell 1-form kernel

Tier 3 (unbounded); partial.

The strongest result in the notes archive, and the first to survive a full-proof audit: $|G(b, b')| \leq (2/m^2) \exp(-\eta \text{ dist})$ for $G = (m^2 + \alpha d_1^* d_1)^{-1}$ on lattice 1-forms of a finite periodic lattice, with $\eta = \text{arcosh}(1 + m^2/(2\alpha C)) = 2 \text{arsinh}(m/(2\sqrt{\alpha C}))$ – an $O(m)$ exponent where naive resolvent bounds give $O(m^2)$. RE-ANCHORED after the audit: the canonical constant C is the WEIGHT-PROFILE row-sum $C_{\text{bdy}}(M_1)$ of Def A.9.4 (Appendix H, Prop H.4.3); the proved...

Ledger status. Appendix H audited in full 2026-08-22, proposition by proposition: the proof is sound end to end, and the no-factor-2 perturbation bound (Lemma H.3.3) is correctly derived and numerically confirmed with near-tight margin (≤ 0.97). Remaining open: the standing T2 sweep at several (m, α) , the H.3.3 mechanism check, the arsinh-form float guard, and the deferred upstream Prop A.9.5.

G22 — Coercivity via Cartan alignment

Tier 3 (unbounded); open.

The notes program’s crux conjecture: on the rough set (average plaquette energy $\geq \epsilon$), configurations far from Cartan alignment have force norm bounded below, volume-uniformly. The original 6-constraints-vs-3-DOF counting argument is refuted (stationarity at a link is 3 equations, and three orthogonal antipodal pairs sum to zero with no common Cartan direction – a registered FINDING check), so whatever truth the conjecture has must come from the gauge-coupled structure across links. Falsifier, required by this register’s own discipline: exhibit a configuration with $B_{\text{avg}} \geq \dots$

Ledger status. The stated finite matrix model has Simon’s transverse confinement bound $H_m \geq T/2 + g \sqrt{m-1} \sum |A_i|$, finite-dimensional compact resolvent and a raw ground-energy floor. These results are retained with their exact scalar split. WILSON_G19_REVIEW identifies the remaining derivation: realize this comparison for the actual gauge-coupled physical operator and prove volume-uniform constants after vacuum subtraction.

Live routes.

- **identify the transverse quantum form and uniform constants for the actual gauge-coupled physical operator.** Identify the actual physical transverse form and retained kinetic allocation. The cited finite-model proof uses finite-dimensional Rellich compactness and bounds E_0 before subtraction; it supplies neither this operator identification nor uniform vacuum-complement coercivity for the coupled lattice.

Completed routes (1). derive transverse confinement along classical flat directions in the stated finite matrix model.

G23 — The OS bridge with uniform constants

Tier 3 (unbounded); open.

Historical notes-program spine as a conditional chain: local curvature $\rightarrow$ LSI $\rightarrow$ diffusion gap $\rightarrow$ exponential clustering $\rightarrow$ OS reconstruction $\rightarrow$ Hamiltonian mass gap, with two named holes that no archive document closes. (1) Uniformity: every implication holds at fixed cutoff and volume once a curvature floor ρ_0 is granted; nothing supplies ρ_0 uniform in a and volume.

Ledger status. The actual fixed-scale Wilson transfer and complete physical odd source band remain established. The exact history theorem gives an isometry and transfer intertwiner for a true reflection/time-compatible pushforward. Gaussian observability now proves that its range is a history-frequency subspace, not an equal-time coordinate count; an arbitrarily small observed residue can make the whole physical two-mode $SU(2)$ history space visible.

Live routes.

- **Transport the nonlinear excess through a compatible actual generated-history hierarchy.** Use the actual full fast inverse and bounded-mean energy input in the complete nonlinear generated coefficient, retaining the harmonic witness and spatial cancellations. Establish uniform interacting remainder and complete source/history matching with a common physical clock. The chosen local source chart need not generate the whole matrix observation sigma algebra;
- **Compare the actual generated history dynamics with a controlled physical low-energy model.** The actual history map, jointly invariant algebra, generated measure, memory and physical clock remain explicit. Literal true-vacuum source ranges now have complete low-window frames and fast floors in additive Wilson models, including common Gauss, and in the regulated full Gaussian boundary model. These equal-time ranges need not be the reducing OS-history range.
- **control the actual physical trial residual after local vacuum cancellation.** WR14–WR16 construct a gauge-invariant trial with a nonnegative shared-link residual, but its global sum is not a centered gap budget. Connected residual and complete boundary-source control are still needed. Independent copies refute a naive extensive residual loss.
- **prove physical conditional assembly and improved coercivity through spatial refinement.** Fixed-spacing substeps are complete: BA20–BA26 Cauchy-Schwarz true-vacuum bound eliminates the multiplicity penalty on single-link partitions ($\kappa \leq 10/229$, gap > 2.8679 epsilon). SC17 bootstrap closes the connected spatial estimate for $\lambda \leq 1/73$, establishing weighted Hessian row $\leq 2/7$ at weight $1025/1024$, projection angle row $\kappa \leq 10658/24923$, $C_{AT} \leq 24923/14265 < 7/4$, and physical gap > 4 epsilon / 3. The rational budgets are machine-certified by the supplied Lean lemmas; *Cannot decide*: G19, G23.

Untried routes.

- **establish tunable physical-time localization on a complete observable class.** Prove the localization and bad-event bounds for every radius with controlled prefactors and complete physical observables. A diffusion gap alone does not identify Euclidean-time spectral decay. A source with an unobserved slow mode falsifies an incomplete-source argument.

Dead routes, and why they died.

- **transfer an isolated physical rotor gap to a fixed finite unreduced boundary frame.**

WR20 embeds the entire single-link electric Dirichlet form using the true patch ground and its Haar marginal. The unreduced boundary gap is at most 3ϵ and the number of modes below a fixed target grows as $\epsilon^{-(3/2)}$. This refutes the fixed finite frame, not a physical gap or assembly with full compact boundary retention. *Cannot decide:* G19, G23.

- **replace physical spatial control by globally positive classical Wilson orbit curvature at weak coupling.** The exact regular SU2 plaquette curvature is $8/3\beta/2$ at $\theta=2\pi/3$. Horizontal geodesics extend the failure to the stated finite cubic lattices. The classical weight is not the physical ground and Wilson flow has an expansion witness $8/7$. *Cannot decide:* G19, G23.
- **obtain a positive physical bound by discarding WR25 local remainders before vacuum subtraction.** VA1-VA8 give an actual connected periodic SU2 Wilson test at $L \geq 3$, $k/\epsilon \geq 32$. The discarded vacuum remainder is at least $15\delta p_0 |E|$ with $p_0 > 0$ independent of volume. A physical mean-zero local phase has expectation of A-F at most $\delta(4-15p_0 |E|)$, negative at large volume. *Cannot decide:* G19, G23.

Completed routes (15). Identify the actual nonreducing Gaussian fast inverse with its energy prior; Identify exact Gaussian endpoint memory and its first chosen-source nonlinear coefficient; Retain exact endpoint memory while comparing the entire additive physical low source window; Determine what the exact harmonic history block observes before assigning an eliminated space; repair negative Wilson curvature and derive the complete compact rotor gap; retain the full compact boundary and prove the actual Wilson-link fast barrier; identify the finite ground transform and retain the full trial residual gap budget; derive the physical spectral-weight bound and optimized localization exponent; align conditional projections with the true Wilson vacuum and establish their local floor; optimize the actual conditional block gap and derive its connected angle and pressure identities; prove actual true-vacuum block assembly in the explicit electric strong-coupling window; derive an all-coupling background magnetic comparison and charged physical-time covariance bound; prove SC17 spatial decay and actual conditional assembly through its explicit strong-coupling threshold; construct the SC17 thermodynamic ground law and its closed physical form with nonzero plaquette overlap; identify actual SC17 thermodynamic physical-time correlations with the closed ground-law semigroup.

Discharged gaps

Finished work stays recorded rather than re-queued. Each entry names what closed it.

- **G18 The spectral bridge at fixed lattice spacing.** Fixed-spacing closure only; continuum source transport remains G19.
- **G2 Regenerate all displayed tables in canonical u.** discharged 2026-08-21 — the conversion statement is now arithmetic (coupling-erratum suite): the printed towers reproduce $4\Delta(3u/2)$ verbatim in $u = \beta/6$ and would be off by exactly $4r$ under the archived Y reading.
- **G24 Derive the shared-link isotropy premise, the one unproved input of the second-order chain.** discharged 2026-08-28, the day it was opened. The premise is not an assumption: it follows from the order-2 Weingarten values.
- **G3 Fourth-order adjudication — decide C_{shp} where it actually lives.** DISCHARGED 2026-09-04 (ADR 0024). C_{shp} lives in the rotation amplitude ρ , and ρ is now decided:

the historical pipeline’s own Stage-3I ledger, the 2026-09-02 cluster assembly and a third engine agree on every cluster of the rotation record but the adjacent-face cube completion, where the historical ledger holds 8 of 24 orderings;

- **G5 Tetrahedral local Haar-resolvent coefficient.** discharged 2026-08-21 — the $r = 2$ instance of the primitive completion law is re-derived exactly at symbolic N (tetrahedral Haar-resolvent suite, `workhouse.cellular`): $c_2 = -8/(N(N^2-1))$, from two temporal histories through the single 4-link intermediate loop, under the same convention that reproduces the sealed-core cube row, the prism 64/24 sector pair, and the pentagonal 120-history count. The falsification test the mobility theorem proposed did NOT kill the coefficient: the $r = w_{\min} - 2 = 2$ allowance carries nonzero primitive Haar-resolvent...

Appendices

A Reproducibility

Every claim in this document that carries a [T0 Lean], [T1 exact] or [T2 numeric] badge can be re-established from a clean checkout in seconds to minutes. This appendix says how, and — more usefully — says what each kind of re-establishment does and does not prove.

A.1 From a clean checkout

```
git clone https://github.com/ats314/WORKHOUSE
cd WORKHOUSE
uv sync --all-extras --frozen
uv run --no-sync workhouse --help
```

Then:

```
make verify      # every registered check: T1 and T2, a few seconds
make lean        # T0: proof-check the Lean core (requires elan)
make status      # the contradiction and gap registers
workhouse verify --tier 1      # only the exact re-derivations
workhouse verify -v           # every tolerance, printed
workhouse verify --only 'h_4^side' # one claim, with its numbers
```

A single claim is the useful unit. `workhouse verify --only` takes a claim name and prints the numbers it establishes, which is what makes a disagreement arguable rather than merely reported. A certification a reader cannot cheaply reproduce is a claim of authority, not evidence, and the design of this repository takes that seriously enough to attach the reproducing command to every row of `CERTIFIED.md`.

A.2 Querying a claim

```
workhouse why C2      # everything recorded about one id
workhouse why t_N     # a constant: value, formalizations, sources
workhouse branches C2 # both sides of a dispute, side by side
```

```

workhouse derive C2 G3 --out f.md # evidence chain, registered edges only
workhouse search '5/612'          # search by exact VALUE, not by name
workhouse lit --for G19           # published work bearing on a claim
workhouse brief G19 --json        # target snapshot with freshness

```

`search` matches by value rather than spelling, which matters because the join keys of this corpus are exact rationals and no natural-language query retrieves 109151/249696. It also carries two warnings a text search cannot: names the register forbids because they regenerate a dissolved contradiction, and names coined in this program that its own source corpus never uses.

A.3 What each tier actually establishes

[T0 Lean] — Lean 4, no sorry.

The statement is checked by the Lean kernel from explicit definitions, with axioms restricted to `propext`, `Classical.choice` and `Quot.sound`. The axiom set is exported and validated rather than asserted: `scripts/export_lean_dependencies.py` builds strictly and records elaborated constants and transitive axioms into `ledger/lean_dependencies.json`, and the loader checks these against declaration records and source fingerprints. Identifiers mentioned in comments are not proof dependencies.

What a T0 badge does *not* establish is importance. Several registered theorems formalize arithmetic evaluations of named rationals. The registry records for each theorem whether it proves an entire named check (`promotes`) or one ingredient of a larger statement (`lean_support`); supporting a statement never silently promotes the whole statement's tier.

[T1 exact] — exact re-derivation.

The statement is recomputed symbolically in exact arithmetic from stated definitions. Corpus rationals are `sympy.Rational`; values the corpus records only as floats carry a `_NUM` suffix and are Python floats. A float that reads as exact is the most dangerous defect available in this repository, and the boundary is guarded by a dedicated test.

[T2 numeric] — numerical agreement.

Float agreement within a tolerance the check itself prints. A T2 pass is that and nothing more. Tolerances are never widened to make a disagreement disappear; a disagreement is registered as an explicit finding that asserts the discrepancy.

[T3 analytic] — no whole-statement machine certification.

The default. An analytic result may be fully proved and still be T3: the badge reports machine coverage, not mathematical status. Status, evidence and tier are three independent fields in `ledger/results.yaml` for exactly this reason.

A.4 The one inference this apparatus does not license

A finite check certifies the statement it checks. It does not certify a general theorem of which that statement is an instance, and the converse also holds: a complete analytic proof is not weakened by absent Lean coverage.

This cuts both ways in practice, and both directions have caused errors here. A suite of exact checks passing at ranks $N = 3, 4, 5, 6$ is not a statement about all N — which is why the symbolic- N engine of §8 exists, and why its degree bound was retracted when one projection turned out to be uncounted. Conversely, a separate Lean library in this program's archive comprises seventy-one

modules with no `sorry` and was found on audit to prove mostly tautologies; a theorem count is not a measure of coverage, and no count in this document should be read as one.

A.5 Regenerating this document

```
python paper/master/generate_apparatus.py
tectonic paper/master/workhouse_master.tex      # the monograph
tectonic paper/master/workhouse_band_paper.tex  # Part I, standalone
```

The generator reads `literature/index.yaml` and `ledger/{gaps,results,theorems,contradictions}.yaml` and writes the bibliography, the external-evidence table, the route register, the results table and the count macros. Every number the prose quotes is one of those macros. If a count here disagrees with the repository, the generator has not been re-run; the repository is right.

B Verification index

Table 2 lists every registered analytic result with its recorded status and evidence level. The two columns are independent fields and are meant to be read against each other: **proven** with **record-backed** evidence means the argument exists and the machine artefact does not.

Of 143 registered results, 140 are recorded **proven**, 1 **conditional**, 1 **disputed** and 1 **falsified**. The Lean tree carries 439 registered theorems with 0 occurrences of `sorry`.

A caution on both counts. 140 out of 143 is a high proportion because a claim is not registered here until its argument is written down; the register measures what has been recorded, not what fraction of the problem is solved. Likewise 439 theorems is a count of declarations, not of mathematical content — see §A, which explains why a theorem count is not a coverage measure and cites a case in this program’s own archive where seventy-one `sorry`-free modules turned out to prove tautologies.

Table 2: Analytic results by recorded status and evidence level. *Status* is what the argument establishes; *evidence* is what artefact backs it. The two are deliberately independent: a result can be **proven** in status and **record-backed** in evidence, meaning the proof exists and the machine artefact does not.

Result	Status	Evidence
ANISOTROPY_THREE_COORDINATE_SHARP_BOUND	proven	analytic
ARCHIVE_CENTER_DEFECT_OBSTRUCTIONS	proven	analytic
ARCHIVE_CONCENTRATION_LOCALIZATION	proven	analytic
ARCHIVE_FINITE_RANGE_INVERSE	proven	analytic
ARCHIVE_GLOBAL_PAIRING_OBSTRUCTION	proven	analytic
ARCHIVE_HORIZONTAL_RESTRICTION_BOUNDARY	proven	analytic
ARCHIVE_LYAPUNOV_PATCHING	proven	analytic
ARCHIVE_REFLECTION_PERMANENCE	proven	analytic
ARCHIVE_SIGNED_SU2_DIFFUSION	proven	analytic
ARCHIVE_SMOOTH_PROFILES	proven	analytic

Result	Status	Evidence
ARCHIVE_SOURCE_TILT_AND_ROOTED_BOUNDS	proven	analytic
ASSEMBLED_Q4_COERCIVITY	proven	analytic
BALABAN_MULTISCALE_CAUCHY_SUMMABILITY	proven	analytic
COMMON_SPACE_CONTRACTIVE_DRIFT	proven	analytic
COMPACT_ROTOR_FAST_GAP	proven	analytic
CONDITIONAL_GRADIENT_REPAIR	proven	analytic
CONDITIONAL_QUANTUM_PATH_COVARIANCE	proven	analytic
CREATOR_PARENT_GAP	proven	analytic
CREATOR_VELOCITY_INVERSION	proven	analytic
DIMENSION_FIVE_OPERATOR_IRRELEVANCE	proven	analytic
DYNAMIC_CONDITIONAL_CUBIC_ENERGY	proven	analytic
EARLIER_B6_RETAINED_KRYLOV_CAMPAIGN	conditional	numerical
EARLIER_CONDITIONAL_SPECTRAL_FLOOR	proven	analytic
EARLIER_CORNER_BLOCKING_REFLECTION_DEFECT	proven	analytic
EARLIER_CUBIC_QUOTIENT_REGULARITY	proven	analytic
EARLIER_D4_DISJOINT_STAPLE_COORDINATES	proven	analytic
EARLIER_DECORATED_HISTORY_MERGING	proven	analytic
EARLIER_DETERMINANT_PARITY_REPAIR	proven	analytic
EARLIER_EQUAL_RAY_IDENTIFICATION	proven	analytic
EARLIER_FINITE_ORDER_SUPPORT	proven	analytic
EARLIER_GRAM_NULL_OPERATOR_QUOTIENT	proven	analytic
EARLIER_ISOLATED_POSITIVE_MATRIX_ATOM	proven	analytic
EARLIER_JOINT_RANK_VOLUME_SCALING	proven	analytic
EARLIER_LOCALIZED_FRAME_OBSTRUCTION	proven	analytic
EARLIER_MOVING_ADJOINT_FORCE_DERIVATIVE	proven	analytic
EARLIER_NESTED_QUOTIENT_REDUCTION	proven	analytic
EARLIER_POSITIVE_SECTOR_PHASE_ISOLATION	proven	analytic
EARLIER_REFLECTION_ADAPTED_BLOCKING	proven	analytic
EARLIER_SHELL_SYMMETRY_RITZ_CONTROL	proven	analytic
EARLIER_SPHERE_EQUAL_COUPLING_COVARIANCE	proven	analytic
EARLIER_SU3_POSITIVITY_BOUNDARIES	proven	analytic
EARLIER_VSU_BOUNDED_DOMAIN_WELLPOSEDNESS	proven	analytic
EXPONENTIAL_SOURCE_TOTALITY	proven	analytic
FESHBACH_RESOLVENT_COMPARISON	proven	analytic
FORM_SCHUR_SCALE_COMPARISON	proven	analytic
FULL_GRAPH_SOURCE_CONGRUENCE	proven	analytic
G17_CORRECTED_KP_POLYMER_STABILITY	proven	analytic
G9_COMBINED_MIXING_SUPPORT	proven	analytic
G9_HERMITIAN_FOLDED_SUPPORT	proven	analytic
G9_LOCAL_SIXTH_NONCANCELLATION	proven	analytic
G9_ONE_FACE_SIXTH	proven	analytic
G9_SIXTH_ORDER_RUR_ACTIVATION	disputed	record-backed
GAUSSIAN_FULL_FAST_GREEN	proven	analytic
GAUSSIAN_OS_OBSERVABILITY	proven	analytic
GAUSSIAN_PATH_ENDPOINT_BASELINE	proven	analytic
GAUSSIAN_QUANTUM_FAST_SOURCES	proven	analytic

Result	Status	Evidence
GAUSSIAN_SCHUR_MEMORY	proven	analytic
GROUND_MARGINAL_SCHUR_SCORE	proven	analytic
HODGE_FESHBACH_SPLITTING	proven	analytic
HODGE_INDUCED_EIGHTH_ORDER_MOMENT	proven	analytic
HODGE_LEAKAGE_ZERO_EXTENSION	proven	analytic
HODGE_RANK_ONE_SECULAR_CUBIC	proven	analytic
INTRINSIC_SCORE_CURRENT	proven	analytic
LITERAL_ENDPOINT_SPECTRAL_COMPARISON	proven	analytic
LOCAL_CLASS_WICK_COEFFICIENTS	proven	analytic
LOCAL_CLASS_WICK_RECURRENCE	proven	analytic
LOCAL_CLASS_WICK_SIGNS	proven	analytic
LOCAL_GRADIENT_EXCITATION_SUPPORT	proven	analytic
MESOSCOPIC_NORMALIZED_SOURCE	proven	analytic
MOVING_TIME_MULTICHANNEL_RATE	proven	analytic
MOVING_TIME_MULTICHANNEL_SHARPNESS	proven	analytic
MOVING_TIME_SPECTRAL_GAP	proven	analytic
ODD_ORDER_CENTRE_PARITY	proven	analytic
ORDERED_CONTOUR_ACTIVITIES	proven	analytic
OS_HISTORY_BLOCK_INTERTWINER	proven	analytic
OS_KERNEL_MOVING_TIME_GAP	proven	analytic
PREDICTABLE_SOURCE_DRIFT	proven	analytic
PROJECTIVE_SOURCE_MARTINGALE	proven	analytic
REVERSE_MARTINGALE_BLOCKING	proven	analytic
RICCATI_RESIDUAL_CERTIFICATE	proven	analytic
ROUGH_GAUGE_COERCIVITY_LOWER_BOUND	proven	analytic
RSM_R_CARRIER_SYMBOL_MASTER_THEOREM	proven	analytic
SELECTED_INVERSE_WITHOUT_DIAGONAL	proven	analytic
SHARP_CUTOFF_SOURCE_GAP_BUDGET	proven	analytic
SOURCE_COMPLEMENT_COMPLETE_WINDOW	proven	analytic
SOURCE_CONGRUENCE_METRIC_TRANSPORT	proven	analytic
SOURCE_CURRENT_BOUNDED_W6	proven	analytic
SOURCE_CURRENT_VARIATIONAL_ENERGY	proven	analytic
SQUARE_FIRST_CURRENT_BUDGET	proven	analytic
SQUARE_FIRST_RESIDUAL_CURRENT	proven	analytic
SQUARE_SCALAR_SOURCE_ODD_SCHUR_VANISHING	proven	analytic
TETRAHEDRAL_FLUX_PROJECTION_BOUNDARY	proven	analytic
TETRAHEDRAL_S4_COMMUTANT_RESOLUTION	proven	analytic
TIER_COLLAPSE_ACTUAL_H4_SUPPORT	proven	analytic
TIER_COLLAPSE_IS_R_DEGREE	falsified	analytic
TRUE_GROUND_CENTER_SCORE_OBSTRUCTION	proven	analytic
UNIFORM_MARKED_CLUSTERING	proven	analytic
UNIVERSAL_CELLULAR_HODGE_SPECTRUM	proven	analytic
W6_ACTUAL_CENTERED_COMPLEMENT_SUPPRESSION	proven	analytic
W6_ANGULAR_REFERENCE_SCORE_BUDGET	proven	analytic
W6_R10_CONDITIONAL_DERIVATIVE	proven	analytic
W6_R10_H4_FROM_H0	proven	analytic

Result	Status	Evidence
W6_R10_PROJECTION_JET_RECURRENCE	proven	analytic
W6_UNIFORM_AGMON_COMPLEMENT_GAP	proven	analytic
WILSON_ACTIVITY_EXTRACTION	proven	analytic
WILSON_ACTUAL_BLOCK_FAST_COMPLEMENT	proven	analytic
WILSON_BLOCK_SCORE_OBSTRUCTION	proven	analytic
WILSON_CARDINALITY_CHART	proven	analytic
WILSON_COEFFICIENT_LOCALITY	proven	analytic
WILSON_COMMON_GAUSS_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_COVARIANT_BLOCK_SOURCES	proven	analytic
WILSON_CREATOR_LIMIT	proven	analytic
WILSON_CUBIC_GROUND_TRANSFER	proven	analytic
WILSON_ENDPOINT_COMPLETE_CLUSTER	proven	analytic
WILSON_ENDPOINT_EQUATION	proven	analytic
WILSON_FINITE_CELL_PHYSICAL_GAP	proven	analytic
WILSON_FIXED_ORDER_CHART	proven	analytic
WILSON_FLAT_BACKGROUND_FAST_SOURCES	proven	analytic
WILSON_GLOBAL_VERTICAL_BARRIER	proven	analytic
WILSON_GROUND_BUNDLE_RELATIVE_FORM	proven	analytic
WILSON_HARMONIC_BOUNDARY	proven	analytic
WILSON_HARMONIC_CUBIC_OBSTRUCTION	proven	analytic
WILSON_INFINITY_PHYSICAL_BAND	proven	analytic
WILSON_KINETIC_WINDOW	proven	analytic
WILSON_LITERAL_COARSE_FORM	proven	analytic
WILSON_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_LOCALIZED_TRUE_GROUND_SCORE	proven	analytic
WILSON_PARENT_GNS	proven	analytic
WILSON_PHYSICAL_SOURCE_RANK_REPAIR	proven	analytic
WILSON_ROOTED_CONTRACTION	proven	analytic
WILSON_SAME_WEIGHT_OBSTRUCTION	proven	analytic
WILSON_SECOND_ORDER_CHART	proven	analytic
WILSON_STRIP_BO_FIRST_TERM	proven	analytic
WILSON_SYMMETRIC_CREATORS	proven	analytic
WILSON_THREE_DIMENSIONAL_HARMONIC_BOUNDARY	proven	analytic
WILSON_TRANSFER_RESOLVENT	proven	analytic
WILSON_TWO_SQUARE_PHYSICAL_SHELLS	proven	analytic
WILSON_TWO_STRIP_PHYSICAL_SPLITTING	proven	analytic
WILSON_UNIFORM_FINITE_SHELL	proven	analytic
WILSON_VACUUM_COMPRESSION	proven	analytic
WILSON_VACUUM_SPECTRAL_FLOW	proven	analytic
WILSON_VERTICAL_FAST_ENERGY	proven	analytic
WILSON_WEIGHTED_ACTIVITIES	proven	analytic

C Provenance

The verification repository is one layer over a larger research archive. Both are needed to read a result’s history, and they answer different questions: the repository says what is currently checked, the archive says what was actually done and when.

C.1 The two layers

The repository (<https://github.com/ats314/WORKHOUSE>) holds the ledgers, the invariant suites, the Lean tree, the generated views (`FRONTIER.md`, `CERTIFIED.md`, `index/`) and the manuscripts. Its generated views are maintained by the verifier commands and are never hand-edited: a test fails if either has drifted, because a generated file that has drifted still reads as current.

The archive holds the primary sources: manuscripts, proof collections, appendices, simulation campaigns, a separate Lean library, reference papers and the raw computational runs. Its organizing rule is that nothing is erased — failed attempts, duplicates, superseded drafts and uncommitted work are all preserved, because deleting a refuted claim destroys the evidence that it was tried.

C.2 Reading a source document

Documents under `theory/`, `corpus-import/` and `settlement/` state what someone believed at the time. They are immutable evidence and are never edited to make a check pass; when a check disagrees with a document, the disagreement is registered as a finding. `theory/SHA256SUMS` pins the contents. `theory/superseded/` holds documents kept for the audit trail, and none of them should be read as current.

This matters more than it usually would, because parts of the corpus were written with AI assistance and some of it is wrong. The repository has already caught a reversed tensor-product identity, a one-ulp transcription error, a tolerance quoted tighter than its own data, and a mechanism proposed here and then retracted. Confidence of phrasing carries no evidential weight in either direction.

C.3 Synthesis documents are not evidence

A specific hazard, recorded because it has cost this program time. Several large synthesis documents in the archive were produced by automated agents summarizing primary sources. Audit found the primary sources honest about their own scope and the synthesis layer systematically stripping their caveats — upgrading conditional statements to proven, attaching Lean support that does not cover the statement it is attached to, and citing a formalization library whose theorems are tautologies.

The rule this produced: cite the primary source, never the synthesis of it, and verify that a claimed anchor resolves rather than only that its hash matches. A correct SHA-256 over a fabricated locator is a failure mode that has actually occurred here.

C.4 A measurement of that hazard

The present edition was assembled with a deliberate test of the effect above, and the result is worth recording because it is quantitative.

Thirteen result clusters of this corpus were summarized independently, each by an agent instructed to extract exact statements with their recorded status, hypotheses and verification tier,

and explicitly told not to upgrade anything the ledger records as conditional. Each summary was then audited by a second agent instructed to refute it against the primary files.

All thirteen summaries were found to overstate in at least one respect. The failure modes recurred:

- a status upgraded past what the ledger records;
- a hypothesis present in the ledger and absent from the summary — most often a periodicity condition, a rank restriction, or a regime bound;
- a verification tier asserted with no check behind it, or with a check that tests a tautology;
- novelty asserted against an index the same summary elsewhere notes is incomplete;
- a cited anchor that does not resolve to the named object, with the surrounding hash correct.

Five of those findings changed statements in this document, and each is named at the point of correction rather than silently applied: §7 on the planar reduction, §6 on the rank scope of the operator splitting, §8 on what the formal proof of the coefficient identity actually checks, §12 on chains versus operators, and §12 again on the overlap exponent.

The lesson for the next agent is not that summarization is useless — the extractions found real material that a direct reading missed, including the strongest external corroboration in §11. It is that a summary of this corpus should be treated as a *lead*, never as a citation, and that the audit step is not optional overhead. It changed one statement in every three that mattered.

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